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RESEARCH ARTICLE
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Sulfur_X: A Model of Sulfur Degassing During Magma Ascent

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Key Points:

- Sulfur_X is a new open-source magma degassing model that accurately predicts the volatile and redox evolution of ascending arc magmas
- Sulfur_X shows that sulfur can start degassing in the lower crust or near-surface, depending on the initial $S^{6+}/\Sigma S$ and H_2O in the melt
- The vapor compositions predicted by Sulfur_X can be used to interpret the CO_2/S_T ratios in high-*T* volcanic gases, an eruption precursor

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract The degassing of CO_2 and S from arc volcanoes is fundamentally important to global climate, eruption forecasting, ore deposits, and the cycling of volatiles through subduction zones. However, all existing thermodynamic/empirical models have difficulties reproducing CO_2-H_2O-S trends observed in melt inclusions and provide widely conflicting results regarding the relationships between pressure and CO_2/SO_2 in the vapor. In this study, we develop an open-source degassing model, Sulfur_X, to track the evolution of S, CO_2, H_2O, and redox states in melt and vapor in ascending mafic-intermediate magma. Sulfur_X describes sulfur degassing by parameterizing experimentally derived sulfur partition coefficients for two equilibria: RxnI. FeS (*m*) + H_2O (*v*) → H_2S (*v*) + FeO (m), and RxnII. $CaSO_4$(m) → SO_2 (*v*) + O_2 (*v*) + CaO (m), based on the sulfur speciation in the melt (*m*) and co-existing vapor (*v*). Sulfur_X is also the first to track the evolution of fO_2 and sulfur and iron redox states accurately in the system using electron balance and equilibrium calculations. Our results show that a typical H_2O-rich (4.5 wt.%) arc magma with high initial $S^{6+}/\Sigma S$ ratio (>0.5) will degas much more (~2/3) of its initial sulfur at high pressures (>200 MPa) than H_2O-poor ocean island basalts with low initial $S^{6+}/\Sigma S$ ratio (<0.1), which will degas very little sulfur until shallow pressures (<50 MPa). The pressure-S relationship in the melt predicted by Sulfur_X provides new insights into interpreting the CO_2/S_T ratio measured in high-T volcanic gases in the run-up to the eruption.

Plain Language Summary Understanding how CO_2 and S are emitted from volcanoes, called degassing, is important in interpreting the CO_2/S_T gas precursors to volcanic eruptions and quantifying the total amount of gases released into the atmosphere that are climatically important. However, existing models show significant discrepancies in predicting the behavior of sulfur during degassing. In this study, we employ a new approach to describe sulfur behavior during magma degassing and develop a new model, Sulfur_X, that successfully reproduces the distinct S, CO_2, and H_2O degassing behavior recorded in melts from different volcanoes. Sulfur_X shows that sulfur can either degas early at high pressure or late at low pressure during magma ascent to the surface, depending on the initial sulfur speciation and H_2O contents in the magma. In addition, sulfur is one of the most commonly measured volcanic gas components used for volcano monitoring. Therefore, the predicted compositional evolution of co-existing vapor by Sulfur_X during magma ascent bears directly on the interpretation of CO_2/S_T ratio measured in high-T volcanic gases and the development of eruption forecast models.

1. Introduction

Degassing of arc magmas is the dominant mechanism through which volatiles (H_2O, C, S, and halogens) stored inside the Earth enter the surface environment, with significant effects on climate and redox evolution. Geochemical quantification of gas emissions is also becoming increasingly critical for understanding how volcanoes prepare to erupt. Given that sulfur can be present in both oxidized and reduced form in the melt and vapor, degassing of sulfur during magma ascent has been invoked as one of the mechanisms that drives redox changes in basaltic magmas (Brounce et al., 2017; Longpré et al., 2017; Moussallam et al., 2016). Currently, three approaches, including volcanic gas observations, microanalysis of volatiles in degassing magmas, and thermodynamic modeling, are commonly used to investigate volcanic degassing. In particular, models of volcanic degassing can link the geochemical records of volcanic gases and melts (e.g., Werner et al., 2020). Thus, the three approaches studied in concert provide the most powerful approach to understanding volcanic degassing.

Melt inclusions (MIs) captured inside crystals at varying depths have the potential to record the magma degassing path as a function of pressure. Modeling equilibrium degassing of CO_2 and S is commonly employed to understand the CO_2/SO_2 ratio in volcanic gases, CO_2 and S contents in mineral-hosted MIs, and degassing depths (Burton et al., 2007; Scaillet & Pichavant, 2003). For example, a sudden increase in the CO_2/SO_2 ratio

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in high temperature volcanic gases has been observed prior to eruptions in several arc volcanoes worldwide (Werner et al., 2019) and the anomaly has been modeled as a surge of deeply sourced CO_2-rich gas, based on the contrasting pressure effects on the solubility of CO_2 and S in silicate melt from existing models (Aiuppa et al., 2016; Burton et al., 2007; de Moor et al., 2016). This interpretation is supported by results that model the complementary change in the volatile contents of MIs during degassing (Scaillet & Pichavant, 2003). However, post-entrapment processes can modify MI compositions (Anderson & Brown, 1993; Danyushevsky et al., 2000; Gaetani et al., 2012; Métrich & Wallace, 2008; L. R. Moore et al., 2015). Notably, recent developments in thermodynamic modeling (L. R. Moore et al., 2018; Rasmussen et al., 2017; Steele-Macinnis et al., 2011; Steele-MacInnis et al., 2017; Wallace et al., 2015), Raman spectroscopy (Aster et al., 2016; Esposito et al., 2012; Hartley et al., 2014; L. R. Moore et al., 2015) and experimental rehomogenization (Mironov et al., 2015; L. R. Moore et al., 2018; Rasmussen et al., 2020) have reconstructed the CO_2 concentration in MIs by taking CO_2 stored in the vapor bubbles present inside MIs into account and demonstrated that vapor bubbles can host up to 90% of the CO_2 in the bulk inclusion. Figure 1 shows the CO_2 and S concentrations from experimentally rehomogenized MIs from Volcán de Fuego in Guatemala (Rasmussen et al., 2020). While the naturally quenched Fuego MIs contain a maximum CO_2 content of ~730 ppm (Lloyd et al., 2013), the rehomogenized inclusions contain up to ~3,300 ppm (Rasmussen et al., 2020) (Figure 1). Although S concentration in the MIs was not changed by rehomogenization, S decreases with decreasing CO_2 along the reconstituted degassing trend, which differs dramatically in shape and magnitude from the untreated MIs (Lloyd et al., 2013).

The S-CO_2 co-degassing trend revealed by the rehomogenized MIs from Fuego (as well as Seguam volcano) by Rasmussen et al. (2020) challenges the conventional understanding that S only degasses at low pressure, after CO_2 is almost entirely degassed from the ascending magma (Lesne et al., 2011; Webster & Botcharnikov, 2011; Witham et al., 2012) and calls into question the interpretation that the increase in CO_2/S in volcanic gas results from an influx of deep, CO_2-rich magma. None of the existing forward degassing models can reproduce the CO_2-S co-variation shown in Figure 1. Moreover, different degassing models predict contrasting CO_2-S degassing paths (Figure 1) with similar initial conditions. All these forward models are based on parameters relating the S concentration or fugacity in the gas and the S concentration in the silicate melt. These parameters include sulfide/sulfate solubility of the silicate melt (Burgisser et al., 2015), sulfide/sulfate capacities of the silicate melt (Moretti & Papale, 2004; Scaillet & Pichavant, 2003), and sulfur partition coefficients between vapor and silicate melt (Witham et al., 2012). However, these parameters were only well calibrated by experiments at pressures lower than 300 MPa (Beermann et al., 2011; Botcharnikov et al., 2011; Lesne et al., 2011; Moune et al., 2008; O'Neill & Mavrogenes, 2002), which might have led to inaccurate extrapolation at high pressures. Furthermore, some of the experiments used for calibrating the existing degassing models could suffer from the formation of sulfide/sulfate solids (Beermann et al., 2011; Botcharnikov et al., 2011) that make it difficult to calculate the sulfur concentration in the vapor from mass balance, from inaccurate estimate of fO_2 (Lesne et al., 2011), and/or from S loss to the capsule (Lesne et al., 2011). All these factors might have caused significant discrepancies among existing degassing models even at pressures of 300 MPa or lower. Moreover, there are more recent experiments constraining sulfur partition coefficients in basalts and andesites (Fiege, Holtz, et al., 2014; Fiege et al., 2015; Gennaro et al., 2020; Masotta et al., 2016; Nash et al., 2019; Zajacz, 2014; Zajacz et al., 2012, 2013) that are not included in any existing model calibration.

Inspired by both the homogenized MI data and high-pressure experimental data, we developed a new degassing model as a Python program (available in GitHub: https://github.com/sdecho/Sulfur_X.git), Sulfur_X, to predict the evolution of S, CO_2, and H_2O in basalt and basaltic andesite melts and co-existing vapor by combining existing COH degassing models (e.g., Iacono-Marziano et al., 2012; Newman & Lowenstern, 2002) with newly developed vapor-melt sulfur partition coefficients. Particularly, the newly developed sulfur partition coefficients are informed by a broader range of experiments compared to previous models and are extrapolated to pressures beyond the existing experiments using thermodynamic properties. This model also tracks redox evolution during closed-system S degassing by coupling S-Fe redox equilibrium in the melt with electron balance calculations as S species partition between melt and vapor. A simplified package of calculating sulfur partition coefficients only is also available (https://github.com/sdecho/Sulfur_X_kdonly).

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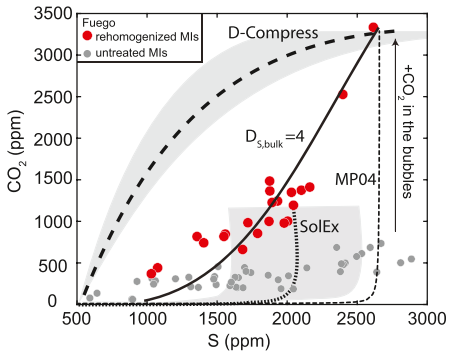

Figure 1. Comparisons of CO_2 and S concentration in experimentally rehomogenized (red circles, Rasmussen et al., 2020) and untreated (gray circles, Lloyd et al., 2013) melt inclusions (MIs) from Fuego (Guatemala) volcanoes with modeled degassing trends from D-Compress (Burgisser et al., 2015), SolEx (Witham et al., 2012), MP04 (Moretti & Papale, 2004), and an empirical model (R2020) fit through the rehomogenized MI data with a constant bulk partition coefficient ($D_S^{\text{non-melt/melt}} = 4$) for sulfur (Rasmussen et al., 2020) between non-melt phases (vapor + silicate minerals) and silicate melt. S uncertainties for D-Compress (+30%, −20%) are reported in Burgisser et al. (2015) at 200 MPa. Since S uncertainties for SolEx are not reported in Witham et al. (2012), the average uncertainties of measured S contents from the experiments that were used to calibrate S degassing (±30%) are plotted. No uncertainty was reported in other models. The empirical fit, D-Compress and MP04 all start with 4.5 wt.% H_2O, 3,300 ppm CO_2, and ~2,600 ppm S. SolEx is forced to start at 400 MPa, the highest pressure can be applied in the model, with 4.3 wt.% H_2O, ~1,100 ppm CO_2 and 2,100 ppm S.

2. Background

2.1. Sulfur Speciation in the Melt and Vapor

Sulfur is a multi-valent element. It exists as S^{2-} and/or S^{6+} in basaltic melt (Jugo et al., 2010; Nash et al., 2019) and as SO_2 and/or H_2S in high-temperature vapor (Holloway, 1987; Figure 2a). When fO_2 of the magma is sufficiently low (<FMQ) or high (>FQM + 2), silicate melts contain dominantly S^{2-} or S^{6+}, respectively. In this case, the maximum sulfur concentration of the magma will be controlled by sulfur content at sulfide (SCSS) or anhydrite saturation (SCAS). SCSS and SCAS for basalts and andesite have been experimentally constrained at different pressures and temperatures (e.g., Smythe et al., 2017; Zajacz & Tsay, 2019). Previous experiments have shown that S^{2-} transitions to S^{6+} within a fO_2 interval between FMQ and FQM + 2 (Figure 2) that also overlaps with the oxygen fugacity range in arc magmas (Jugo et al., 2010). That is, most arc magmas contain both S^{2-} and S^{6+}. Moreover, though minor sulfides have been observed in some arc magmas (Chen et al., 2021; Georgatou et al., 2018; Lloyd et al., 2013; Rose et al., 1978; Zelenski et al., 2018) and ocean island basalts (OIBs, e.g., Wieser et al., 2020, Kilauea) as inclusions in olivines, co-existing with oxides, and/or in the matrix glasses, it is unclear at what stage of magma differentiation sulfide becomes saturated and if it is redissolved in the melt during degassing. Such complexity is commonly resolved by constraints from chalcophile elements (Ranta et al., 2022; Wieser et al., 2020), which is beyond the scope of this study and any other existing degassing model. A current limitation of Sulfur_X is that it does not include the effects of sulfide saturation. We plan to implement this in future development. However, Sulfur_X also has the option to calculate, for reference, the SCSS (Smythe et al., 2017) with a given sulfide composition and SCAS (Zajacz & Tsay, 2019) at each degassing step.

Existing experiments show that S^{2-} and S^{6+} behave differently during degassing (Fiege, Holtz, et al., 2014; Fiege et al., 2015; Zajacz, 2014; Zajacz et al., 2012). Therefore, it is critical to consider sulfur speciation when modeling sulfur degassing, particularly in magmas with both species present. The transition of S^{2-} to S^{6+} in silicate melts as fO_2 increases can be written as follows:

$$S^{2-}_{\text{melt}} + 2O_{2,\text{gas}} = SO^{2-}_{4,\text{melt}} \quad (1)$$

However, there is essentially no free O_2 in magmatic systems. Instead, iron, existing as Fe^{3+} and Fe^{2+} in the silicate melt, is the only other redox-sensitive element in abundance in the system. Therefore, S speciation can be calculated as a function of fO_2 and as a function of the equivalent $Fe^{3+}/\Sigma Fe$ in the melt (e.g., Kress & Carmichael, 1991) (Figure 2a). Nash et al. (2019) combined the oxidation reactions for sulfur and iron to yield the following electron-exchange equilibrium:

$$S^{2-} + 8Fe^{3+} \rightarrow S^{6+} + 8Fe^{2+} \quad (2)$$

To derive the temperature dependence of this reaction more precisely, Nash et al. (2019) rewrote Equation 2 using neutral components in the silicate melt as $FeS + 4Fe_2O_3 = FeSO_4 + 8FeO$ and estimated their thermodynamic properties using solid analogs (e.g., FeS, $FeSO_4$, FeO, Fe_2O_3). However, it remains debatable whether solid analogs can be used to accurately model the temperature dependence of melt species, and whether the temperature dependence of this reaction in silicate melts is as large as estimated by Nash et al. (2019) (O'Neill, 2021; see also Wood et al., 2021). For example, O'Neill's (2021) reassessment of Nash et al.'s thermodynamic modeling suggests a much smaller temperature dependence. Muth and Wallace (2021) adopted O'Neill's smaller temperature dependence and proposed another $Fe^{3+}/\Sigma Fe$ and $S^{6+}/\Sigma S$ relation based on the Fe- and S- X-ray Absorption Near Edge Structure (XANES) spectroscopy data for olivine-hosted MIs from the southern Cascade arc (Muth & Wallace, 2021). However, it is also debated whether the redox states of Fe and S in the MIs could have been modified from that at magmatic temperature during quench (Nash et al., 2019). Recently, O'Neill and Mavrogenes (2022) proposed a method for calculating S^{6+}/S^{2-} in the melt as a func-

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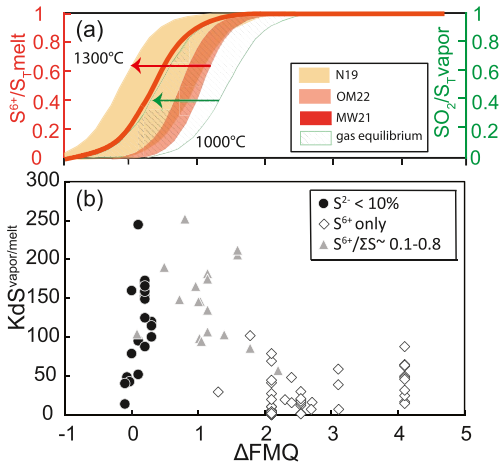

Figure 2. (a) Sulfur speciation in the melt (yellow, light and dark orange) and in the vapor (green) as a function of fO_2 (relative to the fayalite-magnetite-quartz buffer, ΔFMQ, equivalent $Fe^{3+}/\Sigma Fe$ by Kress & Carmichael, 1991) calculated by Nash et al. (2019), O'Neill and Mavrogenes (2022), Muth and Wallace (2021), and gas equilibrium Equation 4. The fO_2 range where S^{2-} transitions to S^{6+} in the melt is predicted to shift one log10 unit lower by the Nash model when temperature increases from 1,050°C to 1,300°C (yellow region) while O'Neill and Mavrogenes (2022) predicts a shift of ~1/3 log10 unit with the same temperature shift (light orange region). Based on the Fe- and S-XANES data for hydrous arc melt inclusions, Muth and Wallace (2021) derived an empirical model shown by the dark orange curve with a very small but reversed temperature dependence (1,300°C at the right edge of the curve, to 1,050°C at the left edge). SO_2/S_T in the vapor phase shifts by 0.83 log unit when temperature increases from 1,050°C to 1,300°C (indicated by arrow in green). All the calculations are carried out at 200 MPa, with a basaltic melt composition and 60 mol% H_2O in the vapor which is reasonable for these conditions. (b) Experimentally determined partition coefficients $\left(kd_S^{v/m}\right)$ of sulfur between vapor and basalt to andesitic melt at pressure between 70 and 500 MPa and temperature of 900°C and 1,250°C (Fiege et al., 2015; Lesne et al., 2011; Masotta et al., 2016; Moune et al., 2008; Zajacz, 2014; Zajacz et al., 2012, 2013) as a function of fO_2 (ΔFMQ). FMQ buffer is calculated at pressure and temperature using the model in Frost (1991) throughout the paper and in Sulfur_X. The experiments are grouped into three categories based on the sulfur speciation in the melt <10% S^{2-} only (black circles), S^{6+} only (open diamonds), and a mixture of S^{2-} and S^{6+} (gray triangles). The S speciation of each experiment was either reported in the original study or calculated using the reported fO_2 (Kress & Carmichael, 1991; Muth & Wallace, 2021; Nash et al., 2019; O'Neill & Mavrogenes, 2022).

tion of temperature, melt composition and oxygen fugacity by combining a newly calibrated sulfate capacity model (O'Neill & Mavrogenes, 2022) with a sulfide capacity model (O'Neill & Mavrogenes, 2002), both of which were calibrated with 1 atm anhydrous experiments. Other experimental studies (Beaudry & Grove, 2019; Matjuschkin et al., 2016; Wilke et al., 2011) demonstrated that water content and pressure could also play a role in sulfur speciation in the melt. Since resolving the Fe-S speciation in the melt is beyond the scope of this study, in Sulfur_X we offer all three currently available models (Figure 2a) from Nash et al. (2019), Muth and Wallace (2021), O'Neill and Mavrogenes (2022) (referred as N19, MW21, and OM22 throughout all figures and figure captions), and the possibility of modifying the Muth and Wallace (2021) model for data approximation if S and Fe speciation data from XANES are available (referred as m_MW21 throughout all figures and figure captions, as shown in the Mauna Kea case below). Since the redox evolution of Sulfur_X can be particularly sensitive to the different $S^{6+}/\Sigma S$ ratios predicted by different models in various regions of T-fO_2 space, we recommend several steps (detailed in the Sulfur_X instructions in Supporting Information S1) when testing Sulfur_X on a new data set.

Equilibrium between sulfur species in the vapor can be described as follows:

$$H_2S + \frac{3}{2}O_2 \rightarrow SO_2 + H_2O \tag{3}$$

Using the thermodynamic properties of the four involved gas components from NIST-JANAF Thermochemical Tables (DOI: https://doi.org/10.18434/T42S31) and adopting a standard state of unit fugacity of the hypothetical ideal gas at a pressure of 1 atm and the temperature of interest for each component, the equilibrium constant of this reaction ($K_{SO_2-H_2S}$) can be written as a function of temperature (T in K):

$$\log_{10}(K_{SO_2-H_2S}(T)) = \log_{10}\left(\frac{fSO_2 fH_2O}{fH_2S fO_2^{3/2}}\right) = -\left(4.1245 - \frac{27,110}{T(K)}\right) \tag{4}$$

where fSO_2, fH_2S, and fH_2O are the fugacities of each gas species. Rearranging Equation 4 shows that fO_2, fH_2O, and temperature are the three important factors dictating the fSO_2/fH_2S ratio in the vapor (Figure 2a). Since most of the previous experiments (except for Lesne et al., 2011) utilized in this study include only S and H_2O, the mole fraction of H_2O in the vapor can be calculated by subtracting the reported sulfur contents in the vapor from 1, assuming only H_2O, H_2S, and/or SO_2 are present in the vapor. Then, fH_2O can be calculated as the product of the mole fraction of water in the vapor, water fugacity coefficient at the given pressure and temperature (Shi & Saxena, 1992), and the total pressure. With knowledge of the fugacity coefficients of SO_2 and H_2S at a given pressure and temperature (Shi & Saxena, 1992), fSO_2/fH_2S can be converted to the ratio of the mole fractions of SO_2 and H_2S. Previous experiments demonstrate that between 650°C and 850°C, 20%–60% of the SO_2 in the fluid can be converted to SO_3 through equilibrium $SO_2 + 0.5O_2 = SO_3$ at pressures from 150 to 300 MPa in Re-ReO_2 buffer (Binder & Keppler, 2011). Increasing temperature and pressure would push the reaction equilibrium to the left side. Extrapolating the experimentally constrained fluid speciation at 150, 200, and 250 MPa to temperatures higher than 1,000°C demonstrates that only at pressures lower than 150 MPa and temperature below 1,100°C, SO_3/S_T could be more than 20% (Figure S1 in Supporting Information S1). Therefore, in our calculations of S species in the vapor, we have ignored the potentially minor amount of SO_3 as well as trace amounts of other S species such as S_2 and COS.

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Table 1
Previous Degassing Models

Model	S Behavior	fO_2	P (MPa)	References
D_Compress	[a]$w_{S^{2-}} = a_{H_2S} \times f_{H_2S}{}^{b_{H_2S}}$	Change	25–200	Burgisser et al. (2015)
	$w_{S^{6+}} = a_{SO_2} \times f_{SO_2}{}^{b_{SO_2}}$			
M&P	[b]$C_{S^{2-}} = w_{S^{2-}} \times (fO_2/fS_2)^{1/2}$	Fixed	0.1	Moretti and Papale (2004)
	$C_{SO_4^{2-}} = w_{S^{6+}} \times fO_2^{-3/2} \times fS_2^{-1/2}$			
SolEx	[c]$kd_S^{v/m} = w_S^{vapor}/w_S^{melt}$	Oxidized only	25–400	Witham et al. (2012)
Empirical models	$kd_S^{v/m} = w_S^{vapor}/w_S^{melt}$	Tailored for one volcano	–	

[a]$w_{S^{2-}}$ and $w_{S^{6+}}$: concentration of S^{2-} and S^{6+} in the melt; f_{H_2S} and f_{SO_2}: fugacity of H_2S and SO_2; a_i and b_i: model constants. [b]$C_{S^{2-}}$ and $C_{SO_4^{2-}}$: sulfide and sulfate capacity; fO_2: oxygen fugacity. [c]w_S^{vapor} and w_S^{melt}: sulfur concentration in the vapor and melt.

2.2. Existing Degassing Models

Previous studies have established computational thermodynamic models to predict the H_2O-CO_2-S evolution in magma and co-existing vapor as a function of T, P, melt composition, and fO_2 (Table 1) and have used the modeled results to interpret volcanic gas (Burton et al., 2007) and MI records (Scaillet & Pichavant, 2003). Figure 3 shows modeled equilibrium degassing paths for H_2O, CO_2, and S in melt (Figures 3a and 3b) and CO_2/SO_2 in co-existing vapor (Figure 3c) calculated using SolEx (Witham et al., 2012), D-Compress (Burgisser et al., 2015) and the degassing model from Moretti and Papale (2004, referred to as MP04) with initial concentrations comparable to those measured in Fuego MIs: ~3,300 ppm CO_2, 2,700 ppm S, and 4.5 wt.% H_2O (with the exception of SolEx, where initial concentrations were lower in S, 2,100 ppm, and CO_2, ~1,100 ppm, due to the model pressure limit of 400 MPa; more details in Figure 3 caption). The empirical fit for the rehomogenized MI data from the Fuego 1974 eruption from Rasmussen et al. (2020) is also plotted for comparison. No solid sulfur-bearing phase (sulfide and sulfate) was included in any of these models. SolEx and MP04 both predict that S decreases significantly in the melt only during the last 150 MPa (Figure 3b). In contrast, D-Compress predicts more rapid S degassing than CO_2 from the melt (Figure 1b) and very low CO_2/SO_2 (<1) in the co-existing vapor along the entire degassing path (Figure 3c). More importantly, no existing forward models can reproduce the S-CO_2 evolution in melt revealed by the rehomogenized olivine-hosted MIs from the Fuego 1,974 eruption (Figure 1; Rasmussen et al., 2020).

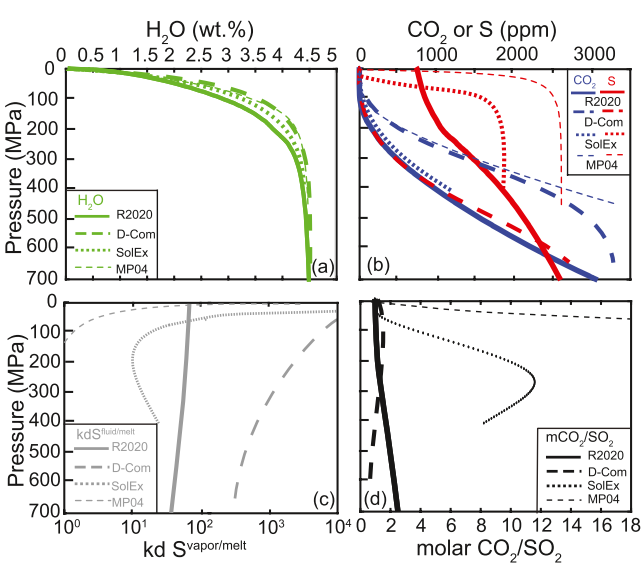

Figure 3. (a) H_2O (green), (b) CO_2 (blue), and S (red) evolution in melt during decompression degassing using D-Compress (Burgisser et al., 2015), SolEx (Witham et al., 2012), MP04 (Moretti & Papale, 2004) and an empirical model that fits Fuego, R2020 (Rasmussen et al., 2020). (c) Calculated partition coefficients of S between fluid and basaltic melt from four models. (d) Molar CO_2/SO_2 ratios in the coexisting vapor for the same three models in (a) and (b). Starting compositions are as in Figure 1.

The discrepancies between the model results from all existing forward models (MP04, SolEx, D-Compress) and the rehomogenized MIs are likely due to the lack of inclusion of more recent experiments conducted at pressures higher than 300 MPa. For example, Moretti and Papale (2004) and Scaillet and Pichavant (2003) employed the concept of sulfate and sulfide capacities to model sulfur behavior, and their models predict high CO_2/SO_2 (>15) in co-existing vapor until very shallow depths (Figure 3c). However, both sulfate and sulfide capacities are only calibrated at 1 atm in dry melts (Boulliung & Wood, 2022; O'Neill & Mavrogenes, 2002, 2022) and the more recent experiments on sulfate capacity were not included in these models (Boulliung & Wood, 2022; O'Neill & Mavrogenes, 2022). The predicted rapid S loss from the melt by D-Compress is accompanied by unreasonably high total S (>50 mol%), high SO_2, H_2S, and S_2 fugacity (discussed further in a later Section 4.1.2) in the vapor at pressures higher than 300 MPa compared to any measured volcanic gas data. These are likely results of the approach taken by D-Compress that the system maintains vapor saturation with a given partial pressure of H_2O, CO_2, and partial pressure different sulfur species are calculated to reach the total pressure (Burgisser et al., 2015). Moreover, the default S dissolution relations employed by D-Compress are calibrated only

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with sulfide/sulfate-saturated experiments (Beermann et al., 2011; Botcharnikov et al., 2011; Moune et al., 2008) at low or high fO_2, but no experiments at intermediate fO_2 that are of most relevance of arc magmas.

Unlike the case for CO_2 and H_2O, silicate melts are never saturated in a pure sulfur vapor or fluid phase (a solid or immiscible melt phase saturates before this state can occur). Thus, simple solubility relationships, as developed for H_2O and CO_2 (Iacono-Marziano et al., 2012; Newman & Lowenstern, 2002), cannot be applied to sulfur degassing. Instead, sulfur partitions between the vapor and melt, where the ratio of S concentration in the vapor to that in the melt can be described by a partition coefficient, $kd_S^{v/m}$, expressed either by ratios of molar concentration ($=XS_{\text{vapor}}/XS_{\text{melt}}$) or of weight concentration ($=cS_{\text{vapor}}/cS_{\text{melt}}$). Therefore, another common method for modeling COHS degassing is to couple an existing COH degassing model (e.g., VolatileCalc) with a sulfur partitioning model (Burton et al., 2007; Gennaro et al., 2020; Johnson et al., 2010; Rasmussen et al., 2020; Sisson & Layne, 1993; Witham et al., 2012). Values for $kd_S^{v/m}$ are usually based on experimental studies (Burton et al., 2007; Gennaro et al., 2020; Witham et al., 2012) or empirically determined from MI data (Johnson et al., 2010; Rasmussen et al., 2020; Sisson & Layne, 1993). For example, the solid curves in Figures 1 and 3 are results of an empirical fit of the rehomogenized MIs by Rasmussen et al. (2020) that combines the COH degassing results from VolatileCalc and a fixed bulk partition coefficient of 4 for sulfur ($D_S^{\text{non-melt/melt}} = kd_S^{v/m} \times \frac{mX_v}{mX_v+mX_{\min}} + kd_s^{\min/m} \times \frac{mX_{\min}}{mX_v+mX_{\min}} = 4$, where both $kd_S^{v/m}$ and $kd_s^{\min/m}$ are ratios of weight percent; m_v and $m_{\min}$ represent mass fraction of vapor and minerals) between non-melt phases (vapor + minerals) and silicate melt. Unlike most of the degassing models of this kind and other forward thermodynamic models that only consider melt and vapor in the system, Rasmussen et al. (2020) includes crystallization and predicts that $kd_S^{v/m}\left(= c_S^{\text{vapor}}/c_S^{\text{melt}}\right)$ increases from 20 to 60 as melt fraction decreases from 100% to 40% along the coupled decompression, degassing, and differentiation path. We use this empirical fit for Fuego when comparing model degassing calculations with Sulfur-X to the S-CO_2 values in the rehomogenized MIs (Section 4.1.2).

Existing experiments show that $kd_S^{v/m}$ is a function of temperature, pressure, melt composition, and oxygen fugacity (Webster & Botcharnikov, 2011). Except for SolEx, $kd_S^{v/m}$ in empirical models (Johnson et al., 2010; Rasmussen et al., 2020; Sisson & Layne, 1993) is tailored to fit the MI data from a specific volcano and cannot be easily applied to other systems. The forward model SolEx (Witham et al., 2012) also fails to reproduce the CO_2-S degassing trend in Figure 1, likely due to several issues with the experiments used to parameterize the model's sulfur partition coefficients (Lesne et al., 2011) that are discussed in Section 2.3. We also note here that all current experiments constrain sulfur partition coefficients by mass balance, which is only valid if the experimental assembly remains a closed system and all sulfur-bearing phases are accounted for. Therefore, other factors that could have compromised the experimentally constrained sulfur partition coefficient include significant H_2 diffusion through capsules, S loss to the capsules, and unquantified sulfide/sulfate crystallization (Section 2.3).

The evolution of fO_2 during magma degassing also represents a challenge to most existing models. Empirical models employ bulk sulfur partition coefficients and do not consider sulfur speciation or redox change, meaning they can only be applied to systems with similar fO_2. For example, sulfur partition coefficients implemented in SolEx were empirically parameterized from experiments with fO_2 between NNO + 1.6 and NNO + 2.1 (Witham et al., 2012). The Moretti and Papale (2004) model considers both SO_2 and H_2S in the gas at a given fO_2 using the gas equilibrium reaction described in Equation 3 and then uses the calculated fSO_2 and fH_2S and sulfate and sulfide capacity to calculate the quantity of S^{6+} and S^{2-} in the melt. However, the lack of experiments constraining the sulfate and sulfide capacity at pressures higher than 1 atm can compromise the S^{6+} and S^{2-} calculations (O'Neill & Mavrogenes, 2002, 2022). Also, the model can only be run at a fixed fO_2 along the degassing path. D-Compress considers the evolution of redox states of the system based on multiple equilibria involving C-, O-, H-, and S-bearing species in the vapor and the ferrous and ferric iron in the magma (Burgisser et al., 2015). This model has been used to describe redox evolution in different magmatic systems (Burgisser & Scaillet, 2007; Longpré et al., 2017; Moussallam et al., 2016) and atmosphere evolution at planetary surfaces (Gaillard & Scaillet, 2014; Gaillard et al., 2011; Liggins et al., 2020). However, because it does not consider S-Fe redox equilibrium in the silicate melt (Equation 3), D-Compress may not accurately predict fO_2 evolution. A recent degassing model from Hughes et al. (2022) tracks the fO_2 evolution during degassing with a similar strategy as D-Compress and calculates S^{6+} and S^{2-} in the melt using fSO_2 and fH_2S, and sulfate and sulfide capacities that are calibrated with anhydrous experiments at 1 atm (Boulliung & Wood, 2022; O'Neill & Mavrogenes, 2002, 2022). Overall, fO_2 evolution in previous models is entirely controlled by the equilibrium reactions among multiple species in the vapor. $Fe^{3+}/\Sigma Fe$ in the melt only responds to the calculated fO_2, and $S^{6+}/\Sigma S$ in the melt is determined by fSO_2 and fH_2S, either through solubility laws that are a function of the corresponding species fugacities (Burgisser

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et al., 2015) or sulfate and sulfide capacities (Hughes et al., in review). None of the existing models directly consider the S-Fe redox equilibrium in the melt or feedback from this equilibrium on the fO_2 of the system.

2.3. Previous S Partitioning Experiments

$kd_S^{v/m}$ has been investigated in previous experimental studies of basaltic-andesitic systems under controlled fO_2 from FMQ-1 to FMQ + 4, T of 900°C–1,400°C, and P of 0.1–500 MPa (Fiege, Behrens, et al., 2014; Fiege et al., 2015; Gennaro et al., 2020; Lesne et al., 2011; Masotta et al., 2016; Moune et al., 2008; Nash et al., 2019; O'Neill & Mavrogenes, 2002; Zajacz, 2014; Zajacz et al., 2012, 2013). However, due to the multiple redox states of sulfur in the melt and vapor, the paucity of experiments at pressures higher than 300 MPa, and the technical difficulties in conducting S-bearing hydrothermal experiments (Webster & Botcharnikov, 2011), there is no existing thermodynamic or empirical model to predict $kd_S^{v/m}$ in the basaltic-andesitic system across reduced and oxidized conditions. This is the goal of this paper.

Existing experiments conducted at pressures of 70–500 MPa and temperatures of 900°C–1,250°C show complex variations in $kd_S^{v/m}$ with increasing fO_2 (Figure 2b). In experiments dominated by S^{2-} (>90%), $kd_S^{v/m}$ increases with increasing fO_2 from <FMQ to FMQ + ~0.3, while in experiments with mostly S^{6+}, $kd_S^{v/m}$ decreases with increasing fO_2 from FMQ + 1.5 to FMQ + 4. Experiments plotted in Figure 2b include different melt compositions (within the range of basalts and andesites) and variable pressures and temperatures, leading to scatter at a given fO_2. Previous experiments have shown that S bonds with different cations in the silicate melt depending on its speciation (S^{2-} or S^{6+}), hence, $kd_S^{v/m}$ is affected by various compositional factors at different fO_2 (Fiege et al., 2015; Masotta et al., 2016; Zajacz, 2014). Under oxidized conditions, where S exists dominantly as SO_4^{2-} and bonds with Ca^{2+}, K^+, and Na^+, $kd_S^{v/m}$ is affected by NBO/T (non-bridging oxygen) and the abundance of cations (Ca^{2+}, K^+, and Na^+) over what is required to charge balance Al–Si complexes in the silicate melt structure (Masotta et al., 2016; Zajacz, 2014). At reduced conditions, when S^{2-} is dominant in the melt and bonds with Fe^{2+}, $kd_S^{v/m}$ decreases as $Fe^{2+}O$ increases (Fiege et al., 2015). Thus, there is no universal model for $kd_S^{v/m}$ that can capture the behavior of the different melt and vapor species of S in all the experiments across the range of fO_2 and melt compositions.

Pressure is another important factor. Recent studies have used multiple approaches to reconstruct the CO_2 concentration in MIs that have lost CO_2 to shrinkage bubbles (Aster et al., 2016; Esposito et al., 2012; Hartley et al., 2014; Mironov et al., 2015; L. R. Moore et al., 2015, 2018; Rasmussen et al., 2017, 2020; Steele-Macinnis et al., 2011; Steele-MacInnis et al., 2017; Wallace et al., 2015). The reconstructions not only dramatically increase estimates for the CO_2 concentration in the MIs (e.g., Figure 1) but also increase the estimates of the depths where arc magma likely starts degassing due to the higher pressures required for higher CO_2 melt concentrations. In particular, the empirical fit of the Fuego rehomogenized MIs from Rasmussen et al. (2020) shows that almost half of the sulfur could have already degassed at pressures >300 MPa. Most experiments constraining $kd_S^{v/m}$, however, have been conducted at pressure of 300 MPa or lower (Fiege, Behrens, et al., 2014; Masotta et al., 2016; Moune et al., 2008; Nash et al., 2019; O'Neill & Mavrogenes, 2002; Zajacz et al., 2012). Zajacz (2014) conducted experiments at 500 MPa to constrain the effect of melt composition on $kd_S^{v/m}$. However, the experiments were performed using synthetic melt compositions with extremely high Na_2O (>10 wt.%), CaO (>20 wt.%), or K_2O (>5wt.%), which makes them inappropriate to directly apply to terrestrial basaltic melts. Lesne et al. (2011) conducted a series of partitioning experiments involving melt and COHS vapor from 50 to 400 MPa. However, large uncertainties in mass balance calculations caused by minimal S partitioning to the fluid in some experiments, significant S loss to the AuPd capsules in others, as well as underestimates of fO_2, and hydrogen loss through the capsule wall could have compromised the derived $kd_S^{v/m}$. For example, experiments on Stromboli compositions at pressures higher than 150 MPa from Lesne et al. (2011) showed almost no sulfur loss to the vapor phase within the S analytical errors of EPMA. On the contrary, experiments on a Masaya volcano melt composition displayed an S increase in the melt as pressure decreased from 400 to 150 MPa (Figure 3d in Lesne et al., 2011) and then a decrease in the melt as pressure continued to decrease. The increase in the S concentration of the melt from 400 to 150 MPa is accompanied by a decrease in run duration, which is consistent with increasing sulfur loss to the AuPd as run duration increases. Lastly, as pointed out by Burgisser et al. (2015), the fO_2 in the experiments from Lesne et al. (2011) might be much lower than what was reported due to Fe reduction induced by hydrogen loss through the capsule wall. The more reduced system could also have been responsible for S loss to the AuPd capsule as S^{2-}. All three scenarios hinder the mass balance calculations of sulfur content in the vapor. As a result, $kd_S^{v/m}$ derived

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from mass balance calculation by Lesne et al. (2011) have considerable uncertainty, and we did not include these experiments in the model calibration. Since the sulfur degassing model in SolEx is exclusively derived from the experiments of Lesne et al. (2011), it is not surprising that SolEx does not reproduce the CO_2-S degassing trend demonstrated by rehomogenized MIs.

Formation of sulfide or sulfate minerals can also compromise the experimentally constrained $kd_S^{v/m}$. Even trace amounts of sulfide or sulfate precipitation could account for a non-negligible amount of sulfur. Such sulfur loss is difficult to quantify and in mass balance calculations gives the appearance that there is more S loss to vapor than actually occurred. In this case, the derived $kd_S^{v/m}$ can only be considered a maximum value (Gennaro et al., 2020) and therefore is not ideal to use for model calibration.

3. A New Degassing Model: Sulfur_X

The approach adopted in this study combines existing COH degassing models (Iacono-Marziano et al., 2012; Newman & Lowenstern, 2002) with sulfur partition coefficients to predict the evolution of S, CO_2, and H_2O in basaltic melt and co-existing vapor. To tackle the challenges of different sulfur species, we consider two sulfur degassing reactions and develop predictive models of $kd_S^{v/m}$ for each reaction guided by relevant published experimental data. The consideration of $kd_S^{v/m}$ for each reaction also allows us to estimate thermodynamic properties of components in the reactions using solid analogs, both as a guide to empirical fitting of the experimental data and to interpolate and extrapolate the pressure and temperature effects beyond the *P-T* range of existing experiments.

3.1. Two $kd_S^{v/m}$

Sulfur can exist in the melt as S^{2-} and S^{6+} (Jugo et al., 2010), and as SO_2 and H_2S in the co-existing vapor. Figure 2a shows S speciation in the melt and in the vapor as a function of fO_2 with temperature between 1,050°C and 1,300°C (Muth & Wallace, 2021; Nash et al., 2019; O'Neill & Mavrogenes, 2022) with other conditions the same (see figure caption). We tackle this complexity by considering $kd_S^{v/m}$ for different degassing reactions individually and calibrate each $kd_S^{v/m}$ using subsets of experimental data conducted under conditions where one of the melt species and one of the vapor species dominate. When the magma is reduced (<FMQ), S^{2-} is the dominant melt species and H_2S is the dominant vapor species. In this case, the sulfur degassing reaction can be described as

$$\mathrm{FeS_{melt}} + \mathrm{H_2O_{vapor}} = \mathrm{H_2S_{vapor}} + \mathrm{FeO_{melt}} \tag{RxnI5}$$

Fe^{2+} is included in this reaction because previous studies have demonstrated that S^{2-} preferably bonds with Fe^{2+} in the silicate melt (Ding et al., 2014; Haughton et al., 1974; Righter et al., 2009). When the magma is very oxidized (>FMQ+2.5), S^{6+} and SO_2 are the dominant sulfur species in the melt and vapor, respectively. In this case, the sulfur degassing reaction can be written as

$$\mathrm{CaSO_{4,melt}} = \mathrm{SO_{2,vapor}} + \mathrm{CaO_{melt}} + 0.5\mathrm{O_{2,vapor}} \tag{RxnII6}$$

We do not assume that SO_4^{2-} is only associated with Ca^{2+}, but we use Ca^{2+} here as an example of one of the important cations bonding with SO_4^{2-} in the silicate melt (Moretti & Ottonello, 2005; O'Neill, 2021; Zajacz, 2014). We note that similar expressions could be written involving other important cations such as K^+ and Na^+, and we include the effect of all major cations in the parameterizations below.

Within the range of fO_2 relevant to arc magmas, between FMQ and FMQ + 2 (Figure 2a), both S^{2-} and S^{6+} exist in the melt and both SO_2 and H_2S in the vapor, and the transitions from S^{2-} to S^{6+} in the melt and from H_2S to SO_2 in the vapor may not be synchronized at the same fO_2 (Figure 2a). For this intermediate fO_2 where multiple S species exist in the equilibrium melt and vapor, we combine partition coefficients of the two endmember degassing reactions (RxnI and RxnII) into a single $kdS_{\mathrm{combined}}^{v/m}$ by weighting each $kd_S^{v/m}$ by the speciation of sulfur in the melt (Equation 11). We show that this empirical method successfully reproduces sulfur partition coefficients constrained by experiments at high, intermediate, and low fO_2 (Figure 11a).

To explore the factors that affect each reaction, we first write the equilibrium constants for RxnI and RxnII.

$$K_{\mathrm{RxnI}}(P,T) = \frac{f_{\mathrm{H_2S}}\, a_{\mathrm{FeO}}}{a_{\mathrm{FeS}}\, f_{\mathrm{H_2O}}} = K^0_{\mathrm{RxnI}} \exp\left(\frac{-\Delta V_{P_0,T_0}(P-P_0)}{RT} + \frac{-\Delta H_{P_0,T_0}}{R}\left(\frac{1}{T}-\frac{1}{T_0}\right)\right) \tag{7}$$

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$$K_{\mathrm{RxnII}}(P,T) = \frac{f_{\mathrm{SO_2}}\, a_{\mathrm{CaO}} f_{\mathrm{O_2}}^{1/2}}{a_{\mathrm{CaSO_4}}} = K^0_{\mathrm{RxnII}} \exp\left(\frac{-\Delta V_{P_0,T_0}(P-P_0)}{RT} + \frac{-\Delta H_{P_0,T_0}}{R}\left(\frac{1}{T}-\frac{1}{T_0}\right)\right) \tag{8}$$

where f represents fugacity and can be expressed as the product of pressure (P), fugacity coefficient (φ), and the mole fraction (X) (e.g., $f_{\mathrm{SO_2}} = P\,\varphi_{\mathrm{SO_{2,v}}}\, X_{\mathrm{SO_{2,v}}}$); a is the activity of the components in the silicate melt and can be expressed as the product of the activity coefficient (γ) and the mole fraction (X) (e.g., $a_{\mathrm{FeO}} = \gamma_{\mathrm{FeO_{melt}}}\, X_{\mathrm{FeO_{melt}}}$); and K^0, $\Delta V_{P_0,T_0}$, and $\Delta H_{P_0,T_0}$ are the equilibrium constant, molar volume change of the incompressible components in the reaction and enthalpy change of the reaction at reference pressure (P_0) and temperature (T_0) (e.g., see Stolper & Holloway, 1988). The values for $kdS^{v/m}_{\mathrm{RxnI}}$ and $kdS^{v/m}_{\mathrm{RxnII}}$ as molar ratios can be expressed (e.g., $kdS^{v/m}_{\mathrm{RxnI}} = {}^{X_{\mathrm{H_2S_v}}}/_{X_{\mathrm{S^{2-}_{melt}}}}$) by rearranging the equilibrium constant expressions above and assuming that $X_{\mathrm{FeS_{melt}}} = X_{\mathrm{S^{2-}_{melt}}}$ in RxnI, and $X_{\mathrm{CaSO_{4,melt}}} = X_{\mathrm{S^{6+}_{melt}}}$ in RxnII.

$$\ln\left(kdS^{\frac{\text{vapor}}{\text{melt}}}_{\mathrm{RxnI}}\right) = \ln\left(\frac{X_{\mathrm{H_2S_{vapor}}}}{X_{\mathrm{S^{2-}_{melt}}}}\right) = \ln K^0_{\mathrm{RxnI}} + \frac{-\Delta V\,(P-P_0)}{RT} + \frac{-\Delta H_{T_0}}{R}\left(\frac{1}{T}-\frac{1}{T_0}\right) + \ln f_{\mathrm{H_2O}} - \ln X_{\mathrm{FeO}} - \ln\gamma\mathrm{FeO} - \ln\mathrm{P} - \ln\varphi\mathrm{H_2S} + \ln\gamma_{\mathrm{FeS}} \tag{9}$$

$$\ln\left(kdS^{\frac{\text{vapor}}{\text{melt}}}_{\mathrm{RxnII}}\right) = \ln\left(\frac{X_{\mathrm{SO_{2,vapor}}}}{X_{\mathrm{S^{6+}_{melt}}}}\right) = \ln K^0_{\mathrm{RxnII}} + \frac{-\Delta V\,(P-P_0)}{RT} + \frac{-\Delta H_{T_0}}{R}\left(\frac{1}{T}-\frac{1}{T_0}\right) - 0.5\ln f\mathrm{O_2} - \ln\mathrm{P} - \ln\varphi_{\mathrm{SO_2}} + \ln a_{\mathrm{CaO}} + \ln\gamma_{\mathrm{CaSO_4}} \tag{10}$$

In Equations 9 and 10, $f_{\mathrm{H_2O}}$, $f_{\mathrm{O_2}}$ and X_{FeO} are known from the experiments, $\varphi_{\mathrm{H_2S}}$ and $\varphi_{\mathrm{SO_2}}$ are calculated from the equation of states (Shi & Saxena, 1992), and γ_{FeO} is calculated by the empirical model of O'Neill and Eggins (2002). ΔV and ΔH_{T_0} for each reaction can either be regressed from the experiments accordingly (e.g., Stolper & Holloway, 1988) or estimated using the thermodynamic properties of their solid analogs (JANAF), although we note that O'Neill (2021) cautioned against the latter approach. The activity coefficients, γ_{FeS}, $\gamma_{\mathrm{CaSO_4}}$, and γ_{CaO} in silicate melts are not well constrained. Additionally, Fe^{2+} and Ca^{2+} are the only representative cations that bond with S^{2-} and S^{6+} (see further discussion below). The separation of sulfur degassing into these two reactions clearly shows that $f\mathrm{O_2}$ does not have a monotonic effect on $kd^{v/m}_S$, and it confirms the experimental observation that different compositional factors on $kd^{v/m}_S$ are important depending on the degassing reaction that is dominant. Equations 9 and 10 also show that the pressure effects on the kd values are proportional to $-\ln P$, which is dictated by the thermodynamic definition of fugacity. Particularly, the modeled pressure dependence for RxnII agree well with that shown in experiments from 1 atm to 500 MPa (Figure S5 in Supporting Information S1), a broader range of pressure compared to existing models. We note that although we have chosen to empirically calibrate the two $kd^{v/m}_S$ values separately, the final implementation in Sulfur_X uses a single effective $kd^{v/m}_{\mathrm{combined}}$ value weighted by the speciation of sulfur in the melt (Equation 11), and the final model achieves equilibrium in melt and vapor by coupling of melt Fe-S redox equilibrium and vapor SO_2-H_2S equilibrium.

$$kd^{v/m}_{\mathrm{combined}} = \left(\frac{S^{2-}}{S_t}\right)_{\mathrm{melt}} \times kdS^{v/m}_{\mathrm{RxnI}} + \left(\frac{S^{6+}}{S_t}\right)_{\mathrm{melt}} \times kdS^{v/m}_{\mathrm{RxnII}} \tag{11}$$

To derive $kd^{v/m}_S$ for each reaction, we examine existing $kd^{v/m}_S$ experiments on basalts and basaltic andesites and ascribe them as relevant to RxnI or RxnII, depending on the sulfur speciation in the melt and the vapor (Table 2). Experiments reported as sulfide or sulfate saturated (Beermann et al., 2011; Botcharnikov et al., 2011; Gennaro et al., 2020; O'Neill & Mavrogenes, 2002) are not considered so as to minimize the errors involved in the reported $kd^{v/m}_S$. Therefore, the current version of Sulfur_X does not describe a degassing magma that is buffered by a solid or immiscible sulfide or sulfate phase. This is a limitation of our degassing model. Experiments without reliable determination of $f\mathrm{O_2}$ or sulfur speciation (Lesne et al., 2011) are not considered because they cannot be used to constrain the effect of $f\mathrm{O_2}$ on individual $kd^{v/m}_S$. Applying the two criteria above, we use the 64 experiments from Fiege et al. (2015), Moune et al. (2008), Nash et al. (2019), Zajacz (2014), and Zajacz et al. (2012, 2013) to constrain the $kd^{v/m}_S$ for each reaction. 45 of the 64 experimental studies report S speciation in the melt as determined by XANES (Fiege, Behrens, et al., 2014; Nash et al., 2019), or Raman spectroscopy (Zajacz, 2014) whereas 19 experimental studies report only experimental fO_2 (7 from Moune et al., 2008; and 14 from Zajacz et al., 2012; Zajacz et al., 2013). Therefore, S speciation in each experiment is either taken from the direct $S^{6+}/\Sigma S$ measurements or calculated using all three available S-Fe speciation models when no $S^{6+}/\Sigma S$

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Table 2
Experiments for $kd_S^{v/m}$ Calibration

References	N[a]	P (MPa)	T (°C)	fO_2[b]	Composition	SO_2/S_T in the vapor[c]	S^{6+}/S_T in the melt[d]	Experiment methods
$kdS_{RxnI}^{v/m}$ ($FeS_{melt} - H_2S_{vapor}$)								
Fiege et al. (2015)	2	70	1,050	−0.07 −0.03	Basalt	H_2S (+H_2O)	0, 0.11	IHPV, controlled fO_2
Moune et al. (2008)	7	200, 300	1,050	(−0.1) ~0.3	Basalt	H_2S (+H_2O)	0	IHPV, controlled fO_2
Zajacz et al. (2012, 2013)	12	185, 200	1,000, 1,030	0, 0.2	Andesite	H_2S (+H_2O)	0	IHPV, controlled fO_2
$kdS_{RxnII}^{v/m}$ ($CaSO_{4,melt} - SO_{2,vapor}$)								
Fiege et al. (2015)	11	70	1,150, 1,200	4.1	Basalt	SO_2(+H_2O)	1	IHPV, fO_2 controlled
Zajacz (2014)	23	500	1,240	2.1	Synthesized composition with abnormal Na_2O/CaO/MgO	>0.98 SO_2 ($H_2S + H_2O$)	0.98–1	IHPV, fO_2 controlled
Zajacz et al. (2012)	2	200	1,000	2.54	Andesite	SO_2 > 96%	1	IHPV, fO_2 controlled
Nash et al. (2019)	7	0.1	1,300	1.08–1.5	Basalt-dacite	SO_2 (CO, CO_2, S_2, COS)	0.95–1	CO_CO_2_SO_2 gas mixing

[a]N is the number of experiments used for regression. [b]fO_2 is reported in each experimental study and all are recalculated to be relative to the FMQ buffer here. [c]SO_2/S_T in the vapor is calculated using Equation 3 in the main text using the experimental temperature, pressure and fO_2. [d]S^{6+}/S_T in the melt was examined in Fiege et al. (2015), Nash et al. (2019), and Zajacz (2014). The rest is calculated using Nash et al. (2019) and experimental temperature.

measurements are available. For experiments from Moune et al. (2008) and Zajacz et al. (2012, 2013), the O'Neill and Mavrogenes (2022) and Nash et al. (2019) models agree well with each other, predicting up to 10% S^{6+} in the melt, whereas the Muth and Wallace (2021) model predicts 5%–35% S^{6+} in the melt (Figure S2a in Supporting Information S1). These experiments are used to calibrate $kdS_{RxnI}^{v/m}$. The final predicted partition coefficients using the three different S speciation models are all within 30% of the measured partition coefficients, and the differences resulting from the S speciation models are smaller than errors associated with the modeling itself (Figure S2b in Supporting Information S1). Therefore, the choice between Nash et al. (2019), O'Neill and Mavrogenes (2022), or Muth and Wallace (2021) for the calibration experiments that lack direct S speciation measurements has a small effect on the final predicted partition coefficients. Importantly, the calculated $S^{6+}/\Sigma S$ ratios in the experimental glasses are only used to group the experiments for model calibration; the partition coefficients used for calibration are still the original published values. Since none of the experiments used for calibration contain CO_2 in the system, another underlying assumption made here is that sulfur partitioning between basaltic melt and CO_2-rich vapor or H_2O-rich vapor is similar. On the one hand, this assumption is unavoidable due to the lack of high-pressure experiments with H_2O-CO_2-S that meet our criteria; on the other hand, existing experiments on felsic systems at 200 MPa from Binder et al. (2018) have demonstrated that up to 30 mol% CO_2 in the vapor does not affect sulfur partition coefficients at the Ni-NiO and Re-ReO_2 buffers, which are relevant to reactions Ia and II. Moreover, the indirect effect of CO_2-rich vapor lowered fH_2O is still considered in our calibration. Therefore, although more experiments with the three volatiles at high pressures are needed, it may still be reasonable to apply the partition coefficients calibrated on CO_2-free experiments to CO_2-bearing systems at high pressure. The SO_2/H_2S ratio in the vapor for each experiment was calculated from gas equilibrium Equation 3.

With the calculated or measured S speciation in the melt and the calculated sulfur speciation in the vapor, we were able to categorize the experiments as relevant to RxnI or RxnII. For RxnI and RxnII, experiments with more than 90% S^{2-} or S^{6+} in the melt, and with only H_2S or dominantly SO_2 (>90%) in the vapor are used, respectively. Reference temperature and pressure of each reaction were chosen to be the *P-T* conditions at which most of the relevant experiments were done, for simplicity. The composition and *P-T*-fO_2 conditions of experimental studies used to calibrate $kdS_{RxnI}^{v/m}$ and $kdS_{RxnII}^{fl/m}$ are plotted in Figures 4d and 4e, and listed in Table 2. Details of each experiment from these studies are given in Table S1 in Supporting Information S1.

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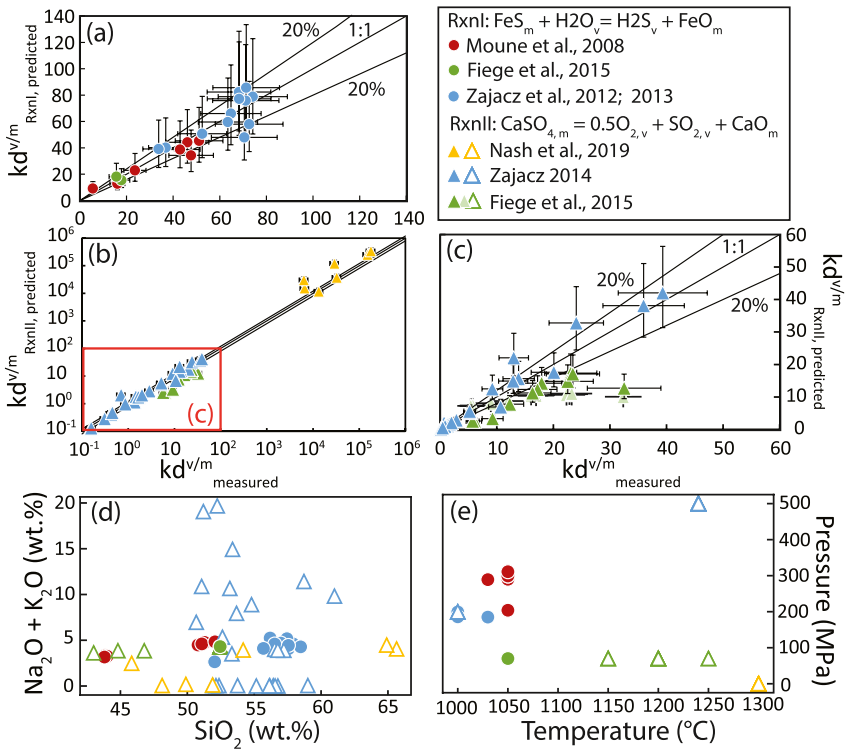

Figure 4. Comparisons between the measured $k_d(S)^{v/m}$ and the predicted $k_d(S)^{v/m}$ for RxnI (a), RxnII (b, c) and melt compositional (d) and *P-T* space (e) of the experiments used to calibrate $kdS^{v/m}_{\mathrm{RxnI}}$ (filled circles) and $kdS^{v/m}_{\mathrm{RxnII}}$ (open triangles). Different symbol colors represent different experimental studies. The error bars of the measured partition coefficients are assigned 20% for RxnI and RxnII. The error bars of the predicted partition coefficients for each reaction are from the inverse regression (see Section S2 in Supporting Information S1), considering errors introduced from the experimental partition coefficients, the estimates of fO_2, and the major element concentrations measured by electron probe.

Experiments within each reaction group span pressure range (Figure 4e) too small to determine statistically meaningful volume changes (ΔV) by regression. Similarly, for RxnI, experiments did not span a wide enough range of temperature to determine enthalpy change (ΔH_{T_0}). Therefore, we used the thermodynamic properties of each component or their solid analogs for estimating the enthalpy changes. ΔH_{T_0} for RxnI is calculated at the relevant value of T_0 using the enthalpies of their solid analogs and the vapor components from JANAF (Figure S3a in Supporting Information S1). Fiege et al. (2015) constrained the temperature dependence of $kd^{v/m}_S$ at oxidized condition (FMQ + 4) with three sets of basaltic compositions from 1,150°C to 1,250°C. Therefore, for RxnII, we adopted a value for ΔH_{T_0} regressed from these T-series experiments (Figure S3b in Supporting Information S1), and the result is of the same order of magnitude as the value calculated using the thermodynamic properties of the solid analogs and vapor components from JANAF (44,828 vs. 32,379 KJ/mol).

In Equation 9, $\Delta V_{\mathrm{RxnI}} = V^{\mathrm{FeO,melt}}_{\mathrm{mole}} - V^{\mathrm{FeS,sulfide\ melt}}_{\mathrm{mole}}$ (Kress et al., 2008; Lange & Carmichael, 1987). We assume that the value for ΔV is independent of T and P. The molar volume change of RxnII (Equation 10), $\Delta V_{\mathrm{RxnII}} = V^{\mathrm{O^{2-},melt}}_{\mathrm{mole}} - V^{\mathrm{SO_4^{2-},melt}}_{\mathrm{mole}}$, is more difficult to assess. We made an estimate based on a comparison with dissolution of CO_2 as CO_3^{2-} in basaltic melts. Stolper and Holloway (1988) described CO_2 solubility using the following reaction:

$$\mathrm{CO_{2,vapor}} + \mathrm{O^{2-}_{melt}} = \mathrm{CO^{2-}_{3,melt}}$$

Using regression of experimental solubility data, they found $\Delta V_r^{0,\mathrm{melt}} = \left(V^{0,\mathrm{melt}}_{\mathrm{CO_3^{2-}}}\right) - \left(V^{0,\mathrm{melt}}_{\mathrm{O^{2-}}}\right) = 33.0 \pm 0.5\ \mathrm{cm^3/mole}$, where $V^{0,\mathrm{melt}}_{\mathrm{O^{2-}}}$ and $V^{0,\mathrm{melt}}_{\mathrm{CO_3^{2-}}}$ are the partial molar volumes

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Table 3
$kd_S^{v/m}$ From This Study

RxnI: $\ln\left(\frac{X\mathrm{H_2S_{vapor}}}{X S_{\mathrm{melt}}}\right) = \ln K^0_{\mathrm{RxnI}} + \frac{-\Delta V\,(P-P_0)}{R\times T} + \frac{-\Delta H_{T_0}}{R}\left(\frac{1}{T}-\frac{1}{T_0}\right) + \ln f\mathrm{H_2O} - \ln X\mathrm{FeO} - \ln\gamma\mathrm{FeO} - \ln P - \ln\varphi\mathrm{H_2S} + AX\mathrm{FeO} + B(X_2\mathrm{O})^2$						
P_0 (MPa)	T_0 (-C)	ΔV ($\mathrm{cm^3/mol}$)	ΔH_{T_0}	$\ln K^0_{\mathrm{RxnI}}$	A	B
100	1,100	−8.85	44,827.65	0.616843 (*0.492902*)	−8.715144 (*4.250800*)	20.844254 (*4.395644*)
RxnII: $\ln\left(\frac{X\mathrm{SO_{2,vapor}}}{X S_{\mathrm{melt}}}\right) = \ln K^0_{\mathrm{RxnII}} + \frac{-\Delta V\,(P-P_0)}{R\times T} + \frac{-\Delta H_{T_0}}{R}\left(\frac{1}{T}-\frac{1}{T_0}\right) - 0.5\times\ln fO_2 - \ln\mathrm{P} - \ln\varphi\mathrm{SO_2} + A(2\mathrm{Ca}-\mathrm{Al}) + B\frac{\mathrm{NBO}}{T} + C(\mathrm{Na}-\mathrm{Al})$						
P_0 (MPa)	T_0 (-C)	ΔV ($\mathrm{cm^3/mol}$)	ΔH_{T_0}	$\ln K^0_{\mathrm{RxnII}}$	A	B	C
200	1,000	−49.5	32,379	−0.255609 (*0.092976*)	−9.888169 (*0.477521*)	−0.970780 (*0.175012*)	−24.763957 (*0.658748*)

Note. Italic represent mole fractions of different compotnents, pressure, temperature, enthalpy, volume and other parameters.

of these species in the melt. We assume that the pressure dependence for RxnII can be approximated using an alternate form of RxnII:

$$\mathrm{SO_{2,vapor}} + \mathrm{O^{2-}_{melt}} + 0.5\mathrm{O_{2,vapor}} = \mathrm{SO^{2-}_{4,melt}}$$

for which $\Delta V_r^{0,\mathrm{melt}} = \left(V^{0,\mathrm{melt}}_{\mathrm{SO_4^{2-}}}\right) - \left(V^{0,\mathrm{melt}}_{\mathrm{O^{2-}}}\right)$. If the volume change for the carbonate dissolution reaction results mainly from the difference of $2\mathrm{O}^{2-}$, then we can estimate the volume change for RxnII, which has a $3\mathrm{O}^{2-}$ difference, as $\Delta V_{\mathrm{RxnII}} = -1.5\times 33 = -49.5\ \mathrm{cm^3/mol}$. We also emphasize that the pressure effects of the vapor components ($f\mathrm{SO_2}$, $f\mathrm{H_2S}$, $f\mathrm{H_2O}$) are derived from the thermodynamic expressions of $kdS^{v/m}_{\mathrm{RxnI}}$, and $kdS^{fl/m}_{\mathrm{RxnII}}$ (-lnP in Equations 9 and 10) and are independent of the experimental data. Reference temperature and pressure, and all the thermodynamic properties at the reference *P-T* used for each reaction are listed in Table 3.

The last terms to be determined are the equilibrium constants for each reaction and the activity coefficients of FeS (γ_{FeS}), $\mathrm{CaSO_4}$ (γ_{FeS}), and CaO (γ_{CaO}) in the melt. Since $\mathrm{S^{2-}}$ bonds mainly with $\mathrm{Fe^{2+}}$, and potentially with $\mathrm{H^+}$ to form $\mathrm{HS^-}$, we considered $\mathrm{X_{FeO}}$ and $\mathrm{X^2_{H_2O}}$ as two compositional terms either affecting γ_{FeS}, where X is the mole fraction calculated on a single cation base. RxnI experiments with 3–6 wt.% water and 5–14 wt.% FeO in the melt were used to regress for $\ln K^0_{\mathrm{RxnI}}$ and for coefficients of $\mathrm{X_{FeO}}$ and $\mathrm{X^2_{H_2O}}$ after moving all the other terms to the left side of Equation 9. The regression is done by the Inverse Iterative Least Squares Method (Menke, 2012a) (details in Supporting Information S1).

For RxnII (Equation 10), the experimental data are insufficient to calibrate expressions for $\gamma_{\mathrm{CaSO_4}}$ and γ_{CaO}, and it is also important to consider the effects of all cations ($\mathrm{Na^+}$, $\mathrm{K^+}$, $\mathrm{Ca^{2+}}$, $\mathrm{Al^{3+}}$) that may complex with $\mathrm{S^{6+}}$ in the melt. Zajacz (2014) investigated different compositional terms that can affect the partition coefficient of sulfur under oxidized conditions relevant for RxnII. Among the terms investigated by Zajacz (2014), excess Ca ($2X_{\mathrm{Ca}} - X_{\mathrm{Al}}$), where X is the mole fraction calculated on a single cation base, excess Na ($X_{\mathrm{Na}} - X_{\mathrm{Al}}$), and the non-bridging oxygen (NBO/*T*) (calculation following Iacono-Marziano et al., 2012) show good correlations with the $kdS^{v/m}_{\mathrm{RxnII}}$ for experiments in the RxnII group after accounting for the effects of $f\mathrm{O_2}$, pressure and temperature ($R^2 = 0.747$, 0.7482, 0.8384 respectively, Figures S4b, S4d, and S4f in Supporting Information S1).Therefore, our approach was to use 2Ca-Al, Na-Al, and NBO/T in empirical compositional terms in order to calibrate a modified form of Equation 10 that describes partitioning of oxidized S between vapor and melt as a function of melt composition, temperature, pressure and $f\mathrm{O_2}$. The coefficients of each compositional term were regressed using the 43 experiments in RxnII group by the Inverse Iterative Least Squares Method (Menke, 2012b) (Supporting Information S1). The final equations for $kdS^{v/m}_{\mathrm{RxnI}}$ and $kdS^{v/m}_{\mathrm{RxnII}}$, the parameters and regressed coefficients for each of the compositional terms, and the equilibrium constants are all listed in Table 3. Table 3 provides uncertainties (in parentheses, one sigma) for each parameter in the $kd_S^{v/m}$ expressions considering errors introduced from the experimentally derived partition coefficients, the estimates of $f\mathrm{O_2}$, and the major element compositions measured by electron probe. More information on error propagation is in the Supporting Information S1. The individual effects of pressure, temperature, oxygen fugacity, and different compositional components on $kdS^{fl/m}_{\mathrm{RxnI}}$ are plotted in Figure S3 in Supporting Information S1; on $kdS^{fl/m}_{\mathrm{RxnII}}$ in Figure S4 in Supporting Information S1.

Figure 4 shows the comparisons between the predicted and the measured partition coefficients for RxnI (Figure 4a), and RxnII (Figures 4b and 4c). Most experiments in RxnI and RxnII can be reproduced by the

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$kdS^{v/m}_{\mathrm{RxnI}}$ and $kdS^{v/m}_{\mathrm{RxnII}}$ models within 20%. The calculated errors for $\ln\left(kdS^{v/m}_{\mathrm{RxnI}}\right)$ regressions are estimated to be ±0.444004, and ln ($kdS^{v/m}_{\mathrm{RxnII}}$) to be ±0.293387, which translate to 40% uncertainties of predicted $kdS^{fl/m}_{\mathrm{RxnI}}$, and 22% of predicted $kdS^{fl/m}_{\mathrm{RxnII}}$, respectively. More details on the error estimation are provided in the Supporting Information S1. Figures 4d and 4e indicate the melt compositions, pressure and temperature range of the experimental data used for $kdS^{v/m}_{\mathrm{RxnI}}$ and $kdS^{v/m}_{\mathrm{RxnII}}$ calibration. We recommend applying our model within the compositional and *P-T* space indicated by Figures 4d and 4e.

$kdS^{v/m}_{\mathrm{RxnI}}$ and $kdS^{v/m}_{\mathrm{RxnII}}$ describe S^{2-} in the melt degassing to H_2S in the vapor, and S^{6+} to SO_2, respectively. Figure 3a shows that these two degassing equilibria are likely to both be important within the transitional fO_2 range (e.g., FMQ to FMQ + 2, Figure 3), and it is not straightforward to constrain the contribution of each reaction to total sulfur degassing in an empirical model. Therefore, we opted for a simple formulation in which an empirical $kd^{v/m}_{\mathrm{combined}}$ is calculated by taking the weighted sum of $kdS^{v/m}_{\mathrm{RxnI}}$ and $kdS^{v/m}_{\mathrm{RxnII}}$ based on the S speciation in the melt, as expressed in Equation 11. For users interested in S partition coefficients only, we offer a simplified python package taking pressure, temperature, fO_2, melt composition and fH_2O as inputs to calculate $kdS^{v/m}_{\mathrm{RxnI}}$, $kdS^{v/m}_{\mathrm{RxnII}}$, $kd^{v/m}_{\mathrm{combined}}$ and the associated errors (https://github.com/sdecho/Sulfur_X_kdonly).

3.2. The Sulfur_X Model: Combining Partition Coefficients With COH Degassing and Tracking of Redox Changes

The full version of the Sulfur_X model to predict the COHS evolution in melt and co-existing vapor during magma ascent is written as python scripts (Supporting Information S1, also available on Github: https://github.com/sdecho/Sulfur_X.git). It takes the new $kd^{v/m}_{\mathrm{combined}}$ expressions (above) and combines them with (a) an existing COH degassing model (b) mass balance of components, (c) electron balance during Fe and S redox changes, and (d) equilibrium between oxidized and reduced Fe and S in the melt and S in the vapor to describe magma and gas evolution during ascent. Particularly, instead of using the $kd^{v/m}_{S}$ tailored to a single volcanic system (Burton et al., 2007; Gennaro et al., 2020; Johnson et al., 2010; Rasmussen et al., 2020; Sisson & Layne, 1993; Witham et al., 2012), we employed the newly developed $kdS^{v/m}_{\mathrm{RxnI}}$ and $kdS^{v/m}_{\mathrm{RxnII}}$ to include the effects of changing *P*, *T*, H_2O, melt composition, fO_2, and sulfur speciation during magma ascent. This model assumes that all the gaseous species are in equilibrium with their counterparts dissolved in the melt. Only isothermal, closed-system degassing is considered at the current stage.

Figure 5 is a flowchart illustrating the structure of this model. The inputs to the model are the initial melt major element composition, initial concentrations of H_2O, CO_2 and S (H_2O_0, $CO_{2,0}$, S_0), initial fO_2 relative to the FMQ buffer, and initial temperature (Figure 5a). The outputs of the model are $kdS^{v/m}_{\mathrm{RxnI}}$, $kdS^{v/m}_{\mathrm{RxnII}}$, and $kdS^{v/m}_{\mathrm{combined}}$, $Fe^{3+}/\Sigma Fe$, $S^{6+}/\Sigma S$, volatile composition of the melt (in wt.%) and the coexisting vapor (both mol% and fugacity), as well as mass fractions of melt, vapor and crystals (if crystallization of silicate minerals is enabled) for each degassing step. The model can be run with or without crystallization. If crystallization is enabled, the major element variations need to be specified as a function of the degree of crystallization and S, H_2O and CO_2 are assumed to be perfectly incompatible. If the major element change is not specified as a function of the degree of crystallization, the default setup for the Fuego 1974 eruption will be used. In the case of Fuego, the degree of crystallization (as indexed to K_2O concentration) changes as H_2O degasses, following the empirical relation constrained by Lloyd et al. (2013). The initial degassing pressure is calculated using H_2O_0 and $CO_{2,0}$ and an existing COH degassing model (the user can choose COH models from either Iacono-Marziano et al., 2012 or Newman & Lowenstern, 2002; Figure 5b). The pressure drops incrementally from the initial pressure to 1 atm, during which S degassing from melt to vapor is calculated using the empirical $kdS^{v/m}_{\mathrm{combined}}$ from Equation 11, and CO_2-H_2O degassing is calculated with the chosen COH degassing model as above and using mass balance at each degassing step. Since there are no experimental constraints for sulfur partition coefficients between 25 MPa and 1 atm, larger uncertainties are naturally involved in the predicted partition coefficients within this low-pressure range, which might cause large excursions in the predicted fO_2 and S contents in the melt. In this case, Sulfur_X provides the options of (a) terminating the model run at pressures higher than 1 atm if degassing at shallow depths is not of interest, (b) running the degassing models multiple times ("monte" function in the script) to obtain the average behavior of S at low pressure, or (c) increasing the $kdS^{v/m}_{\mathrm{combined}}$ by a fixed number at each degassing step once the pressure drops below a certain threshold pressure (e.g., <5 MPa for the case of Mauna Kea discussed later) to stabilize the model run for fO_2 calculation. The user can specify the final pressure where the model run terminates, the fixed increase of the $kdS^{v/m}_{\mathrm{combined}}$ and the threshold pressure (more in the Supporting Information S1).

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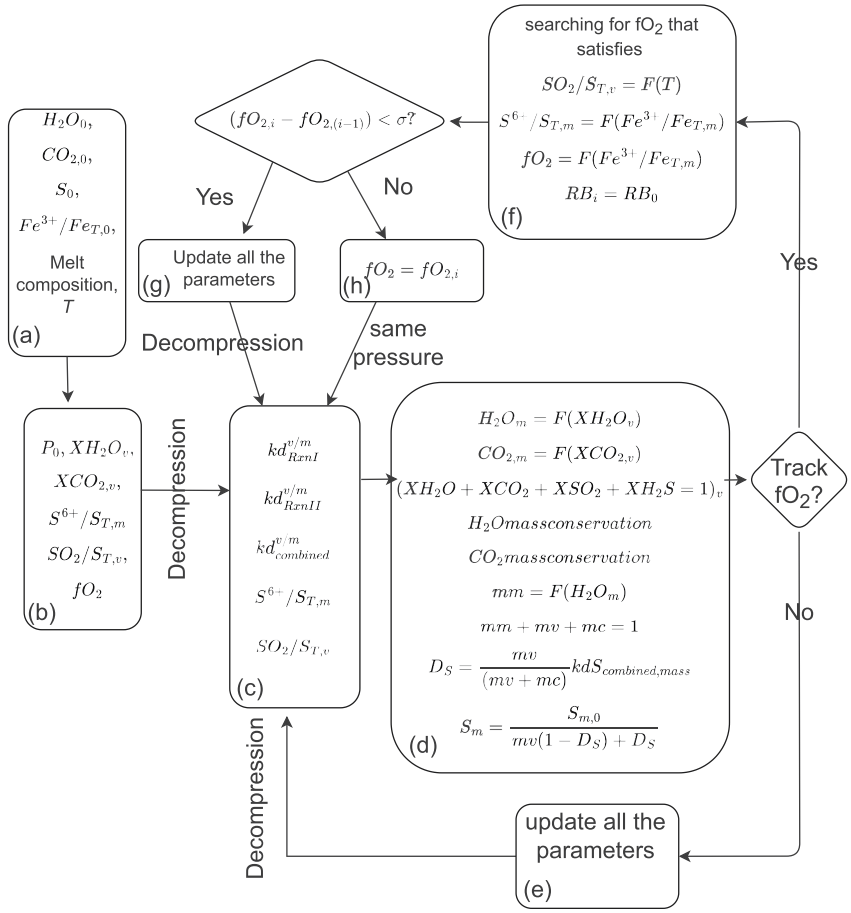

Figure 5. Flow chart of the degassing model. The operations of each box are detailed in the main text. All the required inputs of Sulfur-X are listed in panel (a) and all the outputs are listed in panel (c) and (d). H_2O, CO_2 and S: concentration of each volatile; X: the mole fraction of the components; *mm*, *mv*, and *mc*: mass fractions of melt, vapor and crystals. Subscription 0: the initial condition; *m*: in the melt phase, *v*: in the vapor phase, *i*: degassing step I. *F*(*x*): a function of *x*.

If choosing the linear increase of $kdS^{v/m}_{\text{combined}}$ in the low pressure range, we recommend the use of sulfur partition coefficients constrained from high temperature volcanic gases and degassed glass (Zelenski et al., 2021) as an additional check. For example, the sulfur partition coefficient at 1 atm for most arc and ocean island volcanoes should be on the order of 10^3, expressed in ratios of weight percentage (Zelenski et al., 2021).

The Sulfur_X model has two options for how redox states are handled. In the first, a simplified option, degassing of S species is assumed to have no effect on the redox state of the system. Without considering the redox state changes due to sulfur degassing, fO_2 relative to the FMQ buffer (ΔFMQ) would remain unchanged through the entire degassing path. Because the absolute fO_2 of the FMQ buffer changes with pressure and temperature (Frost, 1991), the absolute fO_2 of the system and $Fe^{3+}/\Sigma Fe$ in the melt are also dependent on decompression and are recalculated as a function of pressure, temperature, and melt composition (Kress & Carmichael, 1991) at every degassing step. At each degassing step, $SO_2/\Sigma S$ in the vapor is calculated with the fO_2 at each current step, the fH_2O from the previous degassing step and the gas equilibrium Equation 2. $kdS^{v/m}_{\text{RxnI}}$, $kdS^{v/m}_{\text{RxnII}}$, and $kd^{v/m}_{\text{combined}}$ are computed using the fH_2O, melt compositions, and sulfur speciation in the melt from the previous degassing step and fO_2 in the vapor at the current step (Figure 5c). The pressure interval between each step can be specified by changing the number of steps; however, the interval needs to be small enough that changes in fH_2O, melt composition, and sulfur speciation in the melt are very small between steps. With $kd^{v/m}_{\text{combined}}$ and the S concentration in the melt from the previous degassing step, total S content, fSO_2, and fH_2S in the vapor can be calculated. Although our approach of using $kd^{v/m}_{\text{combined}}$ is highly empirical, we show below that it successfully reproduces natural data for MIs and submarine glasses. CO_2 ($CO_2^{\text{ppm}}_{\text{melt}}$) and H_2O ($H_2O^{\text{wt.\%}}_{\text{melt}}$) concentrations in the melt are a function of fCO_2 and fH_2O in the co-existing vapor (Iacono-Marziano et al., 2012; Equation 12, 13; Newman &

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Lowenstern, 2002). For example, Equations 12 and 13 include the COH degassing model of Iacono-Marziano et al. (2012). The mass of CO_2 (Equation 14) and H_2O (Equation 15) in the system are conserved during equilibrium degassing. In addition, partial pressures of four volatiles (CO_2, H_2O, SO_2, H_2S) must sum to the total pressure at the current step (Equation 16, Xi_{vapor} represents mole fraction of volatile component i in the vapor). This means that the total pressure calculated from Sulfur_X at any step is larger than the total pressure calculated from the COH degassing models for the equivalent dissolved H_2O and CO_2 concentrations. If crystallization due to H_2O degassing is considered, an empirical linear relation can be used to link the mass fraction of residual melt and the H_2O concentration in the melt (Lloyd et al., 2013, Equation 17). Lastly, the mass fractions of melt (mm), vapor (mv) ± crystals (mc) must sum to 1 (Equation 18) for closed-system degassing. Solving Equations 12–18 together gives the concentrations of H_2O and CO_2 in the melt and vapor, mass fractions of melt (mm) and vapor (mv), and crystals (mc) if crystallization is enabled (Figure 5d).

$$\ln\left(\mathrm{CO_2}^{\mathrm{ppm}}_{\mathrm{melt}}\right) = \sum d_{\mathrm{CO_2},i}x_i + a_{\mathrm{CO_2}}\ln\left(\mathrm{XCO_2}_{\mathrm{vapor}} * P\right) + b_{\mathrm{CO_2}}\mathrm{NBO} + \frac{A_{\mathrm{CO_2}}}{T} + B_{\mathrm{CO_2}} + C_{\mathrm{CO_2}} * P/T \quad (12)$$

$$\ln\left(\mathrm{H_2O}^{\mathrm{wt.\%}}_{\mathrm{melt}}\right) = \sum d_{\mathrm{H_2O},i}x_i + a_{\mathrm{H_2O}}\ln\left(\mathrm{XH_2O}_{\mathrm{vapor}} * P\right) + b_{\mathrm{H_2O}}\mathrm{NBO} + \frac{A_{\mathrm{H_2O}}}{T} + B_{\mathrm{H_2O}} + C_{\mathrm{H_2O}} * P/T \quad (13)$$

$$\mathrm{CO_2}^{\mathrm{ppm}}_{\mathrm{melt}} * mm + \mathrm{CO_2}^{\mathrm{ppm}}_{\mathrm{vapor}} * mv = \mathrm{CO_2}^{\mathrm{ppm}}_{\mathrm{initial}} \quad (14)$$

$$\mathrm{H_2O}^{\mathrm{wt.\%}}_{\mathrm{melt}} * mm + \mathrm{H_2O}^{\mathrm{wt.\%}}_{\mathrm{vapor}} * mv = \mathrm{H_2O}^{\mathrm{wt.\%}}_{\mathrm{initial}} \quad (15)$$

$$X\mathrm{SO_2}_{\mathrm{vapor}} + X\mathrm{H_2S}_{\mathrm{vapor}} + X\mathrm{H_2O}_{\mathrm{vapor}} + X\mathrm{CO_2}_{\mathrm{vapor}} = 1 \quad (16)$$

$$mm = F\left(\mathrm{H_2O}^{\mathrm{wt.\%}}_{\mathrm{melt}}\right) \quad (17)$$

$$mm + mv + mc = 1 \quad (18)$$

With knowledge of the vapor and crystal fraction and $kd^{v/m}_{\text{combined}}$, we calculate the bulk partition coefficient of sulfur between the non-melt phase and melt ($DS^{\text{non}-m/m}_{\text{bulk}}$). Lastly, sulfur concentration in the melt can be calculated using the initial S concentration, bulk S partition coefficient, and mass fraction of the melt (Figure 5d). Without considering redox state changes due to degassing (if redox evolution is disabled), the degassing calculation is finished after updating the CO_2, H_2O, and S concentration in the melt and the fluid, and mass fractions of melt, fluid, and crystals (Figure 5e). Then, the model decompresses to the next pressure step (Figures 5e–5e).

However, as S degasses from S^{2-} in the melt to H_2S and/or SO_2 in the vapor, and from S^{6+} in the melt to SO_2 in the vapor, electrons must exchange among different sulfur species and Fe^{3+} and Fe^{2+} in the melt. Sulfur_X has an option for modeling this electron exchange and resulting redox effects. Other gas species, like CO_2, H_2O, CO, CSO, and H_2 degassing could also cause electron exchange. However, only trace amounts of H_2, CO, and CSO exist in the vapor of arc volcanic systems, and the impacts of these species on the redox state of the system are minor. Therefore, Sulfur_X only considers the redox exchange between S and Fe in the melt and H_2S and SO_2 in the vapor. The model tracks the redox evolution due to sulfur degassing by iteratively searching for a value of fO_2 that satisfies the S and Fe speciation in the melt and S speciation in the vapor after solving for COHS degassing described above.

The target fO_2 must satisfy four conditions (Figure 5f). First, the equilibrium between $SO_2/\Sigma S$ and fO_2 in the vapor (Equation 3) as a function of temperature (Equation 4) and pressure (through the use of fugacity coefficients) should be satisfied. Second, the equilibrium between fO_2 and $Fe^{3+}/\Sigma Fe$ in the melt (Equation 19) from Kress and Carmichael (1991) should also be reached.

$$\ln\left(\frac{X\mathrm{Fe_2O_3}}{X\mathrm{FeO}}_{\mathrm{melt}}\right) = a\ln fO_2 + \frac{b}{T} + C + \sum d_i x_i + e\left[1 - \frac{T0}{T} - \ln\left(\frac{T}{T0}\right)\right] + \frac{fP}{T} + g\left(\frac{T - T0}{T}\right)P + h\frac{P^2}{T} \quad (19)$$

Third, the calculated $Fe^{3+}/\Sigma Fe$ should be in equilibrium with $S^{6+}/\Sigma S$ in the melt (Equation 2), constrained using a relationship such as those proposed by Jugo et al. (2010), Nash et al. (2019), O'Neill and Mavrogenes (2022), or Muth and Wallace (2021) at the given temperature. However, as discussed in Section 2.1, it is still unclear whether the relatively large temperature effect proposed by Nash et al. (2019) is accurate or whether other factors, such as water content and pressure, affect sulfur speciation in the silicate melt. In the case study for Fuego described

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below, given the lack of S and Fe speciation data, we present model results for all three available models (Muth & Wallace, 2021; Nash et al., 2019; O'Neill & Mavrogenes, 2022) and demonstrate that for the temperature and composition of Fuego, modeled S degassing paths are similar regardless of the choice of S-Fe speciation model, although the redox evolution differs in detail (Figure S6 in Supporting Information S1). In the case study of Mauna Kea, we followed the model format of Muth and Wallace (2021) in adopting a smaller temperature dependence and adjusted the equation to fit the S and Fe speciation in Mauna Kea glasses measured by XANES (Brounce et al., 2017). Though the modeled S-CO_2 degassing path for Mauna Kea magma only differs moderately when other S-Fe models are applied, the redox evolution recorded by the glass can only be reproduced by using the modified model that matches the initial S and Fe speciation (Figure S7 in Supporting Information S1). Sulfur_X allows users to adjust the S-Fe redox relationship depending on the volcanic system of interest (more details in the Supporting Information S1).

The last constraint comes from the conservation of the redox budget between Fe and S in the system. The redox budget is defined by Evans (2006) as the number of moles of negative charge that must be added to a sample to reach a reference state:

$$RB = \sum n_i v_i \tag{20}$$

where RB is the redox budget, n_i is the number of moles of an element with a certain valence state, and ν_i is the number of electrons required to take one mol of this element to the reference redox state. For example, the redox budget (RB) of a system with 2 mol of Fe^{2+} and 1 mol of Fe^{3+} and with Fe^{3+} as the reference redox state is $RB = 2\ \text{mol} \times (-1\ \text{electron}) = -2$ mol of electrons. Only Fe and S contribute to the redox budget in our degassing model, and Fe^{3+} and S^{6+} are the reference states. While electrons are exchanged among different sulfur and iron species in the melt, vapor, and potentially crystals, RB of the system of melt, vapor with/without crystals is conserved along the degassing path. Currently, the Sulfur_X model assumes that crystallization does not change the $Fe^{3+}/\Sigma Fe$ ratio in the melt. However, it can be easily modified if information on the crystals is available. Considering the conservation of the RB together with Equations 3, 4, and 19, the degassing model solves for $S^{6+}/\Sigma S_{,\ melt}$, $Fe^{3+}/\Sigma Fe_{\ melt}$, $SO_2/\Sigma S_{vapor}$, and fO_2. With the updated fO_2 (Figure 5h), at the same pressure, the model repeatedly solves for sulfur partition coefficients, COHS degassing, and fO_2 until the fO_2 stops changing within a specified convergence criterion (Figures 5c, 5d, 5f, and 5h). Then, the model updates all the parameters and decompresses to the next pressure step (Figure 5g to Figure 5c). When solving for fO_2, Sulfur_X will sometimes produce a result at a given pressure step that is an outlier, especially at low pressure when $kd^{v/m}_{\text{combined}}$ can be large. These outliers can be minimized by choosing the appropriate model parameters (details in the Supporting Information S1).

4. Application of Sulfur_X to Magma Degassing

We tested Sulfur_X against degassing trends shown by melts from Fuego and Mauna Kea volcano, Hawaii. For Fuego, we used experimentally rehomogenized MIs from the 1974 eruption (Rasmussen et al., 2020). The CO_2 concentrations and the degassing depths of this data set have been reconstructed by rehomogenization experiments, and they display a co-degassing trend of S and CO_2 (Figure 1). These data make a good test case for the new sulfur partition coefficients, particularly their extrapolation to high pressure. We also tested the model using submarine glasses from Mauna Kea volcano (Brounce et al., 2017). The $S^{6+}/\Sigma S$ and $Fe^{3+}/\Sigma Fe$ in the Mauna Kea glasses were measured by XANES and used to test the model predictions of the redox evolution due to sulfur degassing.

4.1. Magma Degassing at Fuego and Comparison to Other Models

For the Fuego 1974 magma degassing model, the initial volatile concentrations are taken from the highest measured values in the MI population (Rasmussen et al., 2020), 4.5 wt.% H_2O, 2,650 ppm S, and 3,300 ppm CO_2 (reconstructed). Lloyd et al. (2013) used V partitioning between the Fuego MIs and their corresponding host olivines to estimate fO_2 values between FMQ + 1.2 for the more mafic phenocrysts (Fo77–78), Fo = XMgO/(XFe^{2+}O + XMgO) and FMQ + 0.76 for Fo73–74 phenocrysts. Thus, the fO_2 in the melt was chosen to be FMQ + 1.2, corresponding to initial $Fe^{3+}/\Sigma Fe$ of ~0.228 at the initial P-T conditions (Kress & Carmichael, 1991). The model is run isothermally at the temperature of 1,030-C (Lloyd et al., 2013). With the initial $Fe^{3+}/\Sigma Fe$

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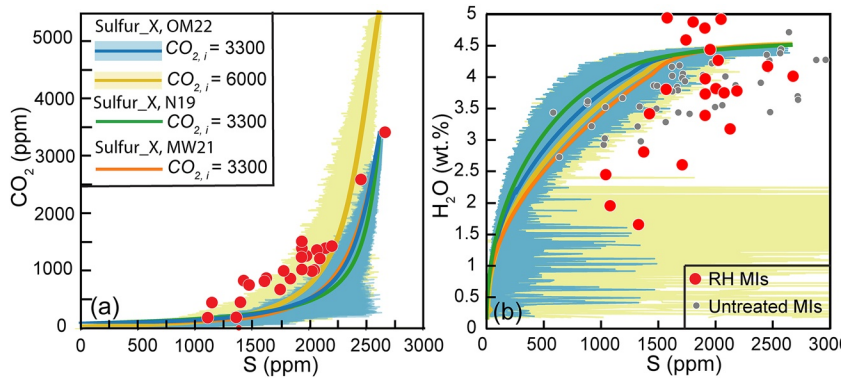

Figure 6. (a) Comparison of CO_2-S concentration between the rehomogenized Fuego melt inclusions (MIs) (Rasmussen et al., 2020) and the Sulfur_X modeled degassing with 3,300 ppm initial CO_2 with speciation models from O'Neill and Mavrogenes (2022, thick blue curve), Muth and Wallace (2021, thick orange curve) and Nash et al. (2019, thick green curve), and 6,000 ppm initial CO_2 (thick yellow curve). (b) Comparison of H_2O-S concentration between the rehomogenized (Rasmussen et al., 2020) and untreated MIs (Lloyd et al., 2013) and the Sulfur-X model results with 3,300 ppm initial CO_2 (same legends as panel a) and 6,000 ppm initial CO_2. Blue and yellow shadings in both panels are results from 500 Monte Carlo simulations, representing the errors of predicted S in the melt. Modeling details are described in the main text.

calculated from the initial fO_2 value (FMQ + 1.2) and temperature, the initial $S^{6+}/\Sigma S$ in the melt is ~0.75, calculated using the O'Neill and Mavrogenes (2022) model. If the Muth and Wallace (2021) or Nash et al. (2019) models are applied, the corresponding initial $S^{6+}/\Sigma S$ in the melt is ~0.85 or 0.46, respectively. Previous studies show that the Fuego melt compositions changed during degassing, from SiO_2 of 48–58 wt.%. as S decreased from 2,650 to 500 ppm (Lloyd et al., 2013; Rasmussen et al., 2020). Therefore, the model includes crystallization, controlled by the empirical relation between H_2O and K_2O concentration (Lloyd et al., 2013) and assuming K_2O is perfectly incompatible. The evolution of other major elements (SiO_2, Al_2O_3, TiO_2, MgO, etc.) along the degassing and crystallization trend is empirically parameterized as a function of the degree of crystallization (Lloyd et al., 2013; Rasmussen et al., 2020). We employed the COH degassing model from Iacono-Marziano et al. (2012) that includes melt composition as a parameter.

Figure 6 compares the rehomogenized (Rasmussen et al., 2020) and naturally quenched (Lloyd et al., 2013) Fuego MIs to the degassing results for S-CO_2 (Figure 6a) and S-H_2O (Figure 6b) from Sulfur_X. Using the O'Neill and Mavrogenes (2022) model (OM22), the solid blue curve shows the results from the initial conditions described above. The light blue shading is the S envelope derived from a Monte Carlo scheme, where the degassing model was run 500 times, with each run employing random numbers generated within 1σ of the predicted $kdS^{v/m}_{\mathrm{RxnI}}$ and $kdS^{v/m}_{\mathrm{RxnII}}$. Large S errors are involved at low pressure (<60 MPa) because of the large predicted $kdS^{v/m}_{\mathrm{RxnI}}$, and $kdS^{v/m}_{\mathrm{RxnII}}$. This output predicted a co-degassing trend of S and CO_2 close to the MI data. Using S-Fe models from Muth and Wallace (2021, MW21) and Nash et al. (2019, N19) produces similar S degassing paths (orange and green curves in Figure 6), with a slightly greater rate of S degassing with the MW21 and slowest S degassing by N19 resulting from different initial $S^{6+}/\Sigma S$. However, even considering the model errors, Sulfur_X predicts higher sulfur at a given CO_2 concentration compared to the Fugeo rehomogenized MIs. One potential explanation for the small mismatch is that the initial CO_2 concentration from the undegassed parental magma could be higher than the highest CO_2 concentration from the rehomogenized experiments. Given that only two rehomogenized MIs with CO_2 higher than 2,000 ppm define the early degassing path, the initial CO_2 from the undegassed parental magmas may be higher than 3,300 ppm. For example, the yellow solid curve and light yellow shading in Figure 6 represent a model run starting with 6,000 ppm CO_2 with the predicted S envelop derived from 500 simulations. For this case, the rehomogenized MI plot mostly within the modeled CO_2-S region. Another potential factor causing the mismatch between the modeled results and the rehomogenized MIs is sulfide saturation. Sulfide inclusions inside MIs, magnetite, and olivine have been observed in products of the 1974 eruption (Lloyd et al., 2013; Rose et al., 1978). However, even a trace amount of sulfide in the crystal assembly would have depleted chalcophile elements, such as Cu and Ag, which were not observed in the K_2O-Cu whole rock data of the 1974 eruption (Figure S8 in Supporting Information S1). Therefore, sulfide saturation through the liquid line of descent can be excluded from Fugeo 1974. A more complicated scenario could be that the magma was sulfide saturated during a certain stage and some of the sulfide was redissolved back to the magma due to sulfur

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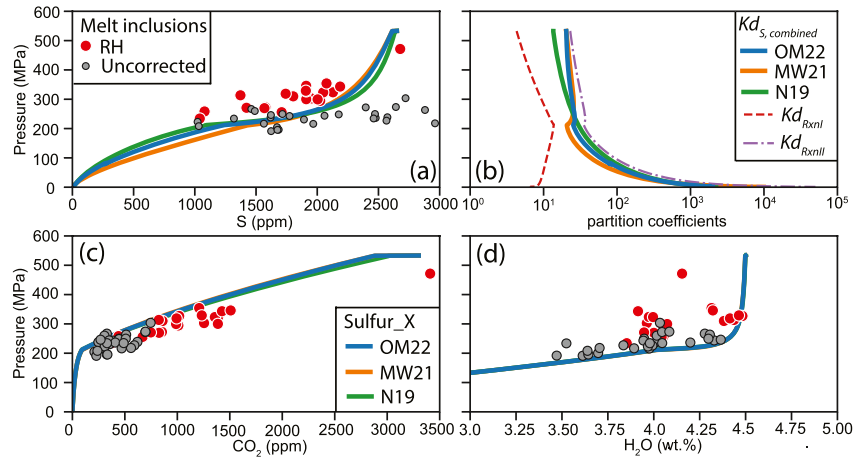

Figure 7. Modeled S (a) $kdS^{v/m}_{\mathrm{RxnI}}$, $kdS^{v/m}_{\mathrm{RxnII}}$, and $kdS^{v/m}_{\mathrm{combined}}$ (b), CO_2 (c) and H_2O (d) change as a function of pressure. In all four panels, thick blue curves are modeled results using OM22, MW21, and N19 speciation models, as shown in Figure 6. The red and purple thick dashed lines are modeled $kdS^{v/m}_{\mathrm{RxnI}}$ and $kdS^{v/m}_{\mathrm{RxnII}}$. Filled red circles represent rehomogenized melt inclusion (MI) data (Rasmussen et al., 2020) and filled gray circles represent untreated MI data (Lloyd et al., 2013). Pressures of both data sets are recalculated using the reported CO_2 concentration, H_2O concentration corrected by K_2O concentration in the MIs, and the COH degassing model by Iacono-Marziano et al. (2012).

degassing or other changes in composition, pressure, and fO_2 change (Botcharnikov et al., 2011; Jugo et al., 2010; Nash et al., 2019). The current version of Sulfur_X includes sulfur degassing only and does not include sulfide saturation. This will be employed in future developments.

Figure 6b shows the modeled S-H_2O degassing trends from low initial CO_2 concentrations with different S-Fe models, and from high initial CO_2 concentrations with the OM22 model. The modeled result demonstrates that significant S degasses (>1,000 ppm) before H_2O degasses (<0.5 wt.%), which is consistent with the naturally quenched MI data from the Fuego 1974 eruption (Lloyd et al., 2013). The rehomogenized MIs show a steeper H_2O decrease as S decreases; however, H_2O concentration in the rehomogenized MIs could be compromised during rehomogenization experiments (Rasmussen et al., 2020).

Figure 7 shows the predicted S, CO_2, and H_2O changes as a function of pressure, closely following the trend in the MI data. Only the results starting with 3,300 ppm initial CO_2 in the melt are shown. Notably, Figure 7a shows that modeled S concentration in the melt decreases from 2,500 to 1,000 ppm between 300 and 200 MPa, similar to the observations from rehomogenized MI.

4.1.1. Changes in $kdS^{v/m}$ Values and Sulfur Speciation as a Function of Pressure

Although the Sulfur_X degassing model describes a CO_2-S path that approximates that defined by the homogenized MIs from Fuego (Figure 6), this result is the sum of complex variations in the $kdS^{v/m}$ values and sulfur species as a function of pressure. Here we explore the variation of each of the individual $kdS^{v/m}_{\mathrm{RxnI}}$ and $kdS^{v/m}_{\mathrm{RxnII}}$ values, and how each contributes to the behavior of bulk sulfur in the melt. Figure 7b shows that $kdS^{v/m}_{\mathrm{RxnII}}$ increases significantly as pressure decreases whereas $kdS^{v/m}_{\mathrm{RxnI}}$ increases slightly with decreasing pressure to 200 MPa then slightly decreases as pressure continues dropping. The significant negative effects of pressure on both partition coefficients come from the $-\ln P$ term in Equations 9 and 10. However, for $kdS^{v/m}_{\mathrm{RxnI}}$, the negative P effect is counteracted by decreasing fH_2O (Figure 8e), which results in decreasing $kdS^{v/m}_{\mathrm{RxnI}}$ at pressure lower than 200 MPa. On the contrary, $kdS^{v/m}_{\mathrm{RxnII}}$ continues increasing by orders of magnitudes, likely due to the decreasing fO_2 (Figure 8d) that increases $kdS^{v/m}_{\mathrm{RxnII}}$ (Equation 10).

Next, we consider how each partition coefficient contributes to $kdS^{v/m}_{\mathrm{combined}}$, based on the sulfur speciation in the melt and vapor. The S speciation in the melt is closely linked to the Fe speciation. $Fe^{3+}/\Sigma Fe$ varies little between 600 and 400 MPa, increases to ~0.29 as pressure decreases to 75 MPa, and then decreases rapidly as pressure continues to drop. The $Fe^{3+}/\Sigma Fe$ change with pressure translates to $S^{6+}/\Sigma S$ (Figure 8c) and ΔFMQ (Figure 8d) showing similar increasing and then decreasing trends. In contrast, SO_2/S_T in the vapor increases from 0.45 up to 0.9 as pressure decreases to the surface, likely due to the decreasing f_{H_2O} (Figure 8e) and the changing fO_2.

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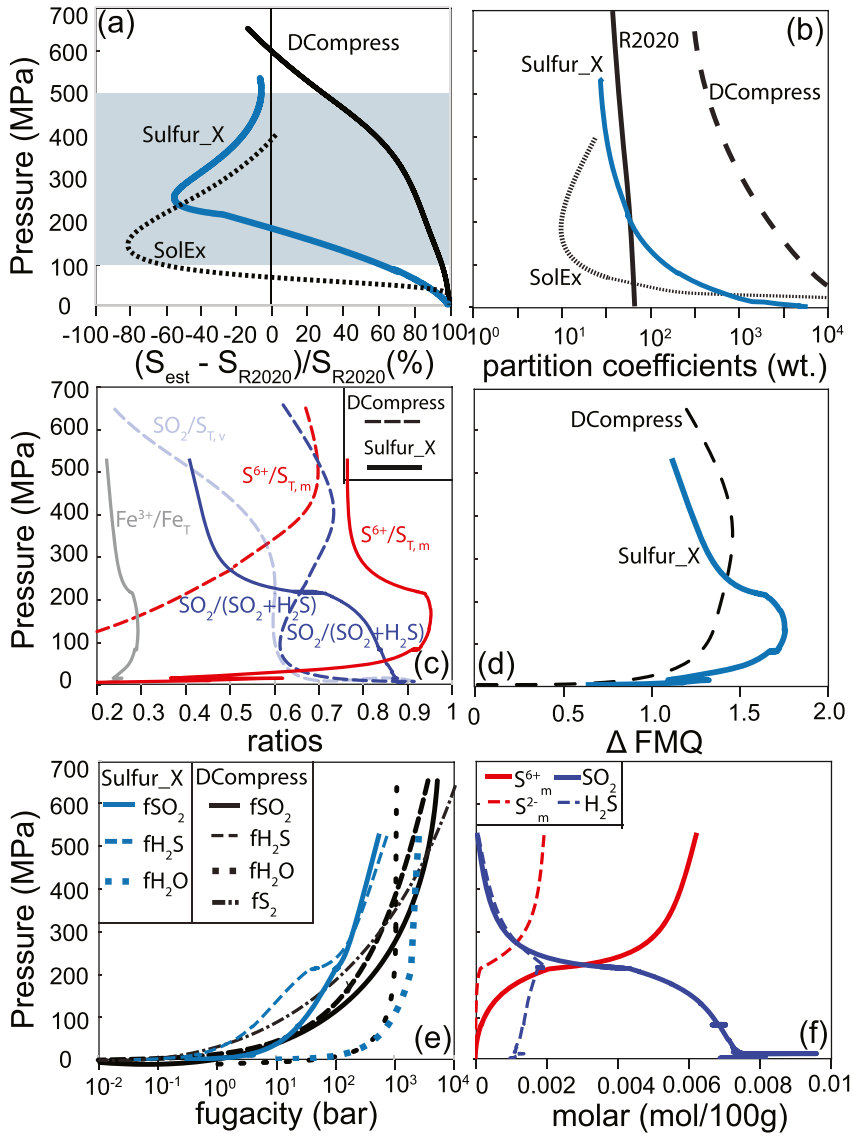

Figure 8. Model results for Fuego with 3,300 ppm initial CO_2, and comparison to previous models, empirical fit (Rasmussen et al., 2020, R2020), SolEx (Witham et al., 2012), and D-Compress (Burgisser et al., 2015). All parameters are plotted against degassing pressure. (a) Comparisons of relative differences ($S_{pred} - S_{R2020}/S_{R2020}$, %) between the S predicted by Sulfur_X (blue curve), D-Compress (long dashed curve) and SolEx (dotted curve) and the empirical fit R2020. (b) Comparisons of apparent sulfur partition coefficients in wt.% from this study (solid blue curve), R2020 (solid black curve), SolEx (dotted curve), and D-Compress (long dashed curve). Apparent sulfur partition coefficients from the previous models are recalculated from $wt\ (S_{vapor})/wt\ (S_{melt})$ in each model. (c) Comparison of calculated S^{6+}/S_T in the melt (red curves) and $SO_2/(SO_2 + H_2S)$ in the vapor (dark blue curves) from this study (solid curves) and D-Compress (long dashed curves). SO_2/S_T (where $S_T = SO_2 + H_2S + S_2$ in moles) from D-Compress is also plotted (light blue long dashed curve). Fe^{3+}/Fe_T in the melt from this study is also shown (solid gray curve). (d) Comparisons between fO_2 (relative to the FMQ buffer) predicted by this study (yellow solid curve) and D-Compress (long dashed curve). (e) Comparison of SO_2 (solid), H_2S (dashed) and H_2O (dotted) fugacity predicted by this study (blue) and D-Compress (black). S_2 fugacity predicted by D-Compress is also plotted (black dash-dot curve). (f) Variations in moles of S^{6+}(red solid curve), S^{2-}(red dashed curve), SO_2 (blue solid curve), and H_2S (blue dashed curve) during degassing, calculated using Sulfur_X with 100 g total weight of the system.

$kdS^{v/m}_{\text{combined}}$ is calculated using $kdS^{v/m}_{\text{RxnI}}$, and $kdS^{v/m}_{\text{RxnII}}$, with each reaction weighted by S speciation in the melt (Equation 11). Because the $S^{6+}/\Sigma S$ of the melt is always >0.7 at pressures higher than 50 MPa, the $kdS^{v/m}_{\text{combined}}$ is always dominated by RxnII. At pressures lower than 50 MPa, the predicted $kdS^{v/m}_{\text{combined}}$ increases rapidly to ~2,000 at 1 atm, almost overlapping with $kdS^{v/m}_{\text{RxnII}}$. The extremely high $kdS^{v/m}_{\text{combined}}$ at low pressures is mainly due to rapid increase of $kdS^{v/m}_{\text{RnxII}}$, resulting from decreasing pressure and fO_2.

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Although the above analysis explains the dominant factors that contribute to $kdS^{v/m}_{\text{combined}}$, it is challenging to tease apart the drivers and responses of S and Fe speciation and fO_2, which are inextricably linked via electron exchange. Sulfur_X captures not only the effect of evolving fO_2 on S degassing but also the electron exchange between Fe and S as a consequence of degassing, depending on the contribution of the two degassing reactions. For example, S^{2-} degassing to H_2S (RxnI) does not cause any electron exchange. During RxnII, one mol of S^{6+} degasses to $S^{4+}O_2$ by receiving 2 mol of electrons, likely donated by the Fe^{2+}. Therefore, S^{6+} degassing to SO_2 (RxnII) increases $Fe^{3+}/\Sigma Fe$. In Fuego's case, each reaction's contribution can be best visualized if we consider the molar change of S^{6+}, S^{2-}, SO_2, and H_2S in a system of 100 g total weight (Figure 8f). At pressures higher than 200 MPa, the increase in moles of H_2S and SO_2 in the vapor phase is almost mirrored by the decrease in S^{2-} and S^{6+} in the melt, suggesting that RxnI and RxnII are the dominant degassing reactions at this stage. As a result of RxnII, $Fe^{3+}/\Sigma Fe$ increases. This oxidizing effect becomes pronounced as the fraction of vapor phase increases, also as the fraction of S^{6+} in the melt increases, responding to the increase in $Fe^{3+}/\Sigma Fe$. Therefore, $kdS^{v/m}_{\text{RxnII}}$ becomes increasingly dominant in the $kdS^{v/m}_{\text{combined}}$. Meanwhile, also responding to the increased $Fe^{3+}/\Sigma Fe$, SO_2/S_T in the vapor increases rapidly, which eventually requires H_2S in the vapor converting to SO_2 (Equation 2). As a result, H_2S in the vapor decreases slightly when both S^{2-} and S^{6+} continue decreasing (Figure 8f). The decrease of H_2S accompanies the significant increase of SO_2 in the vapor that outweighs the decrease of S^{6+} in the melt. The redox change from H_2S to SO_2 in the vapor can efficiently reduce $Fe^{3+}/\Sigma Fe$ in the melt. Therefore, the modeled $Fe^{3+}/\Sigma Fe$ change is the net effect of donating electrons as S^{6+} in the melt degasses to SO_2 (RxnII), and through H_2S conversion to SO_2 in the vapor. The reduction efficiency of the H_2S-SO_2 gas equilibrium outweighs the oxidizing efficiency of RxnII, and results in the rapid reduction of $Fe^{3+}/\Sigma Fe$ during the last stage of degassing (<100 MPa). The predicted S speciation in the melt and in the vapor as well as the redox evolution is different when the MW21 and N19 models are applied (Figure S6 in Supporting Information S1).

4.1.2. Comparison to Other Degassing Models

Figure 8a compares the relative differences in the S concentrations in the melt predicted by Sulfur_X, D-Compress and SolEx, and the empirical fit of the rehomogenized MIs from Rasmussen et al. (2020) as a function of pressure. At pressures between 500 and 100 MPa, where all MIs are entrapped (Figure 7), the relative difference from Sulfur_X is ~±40%, while D-Compress predicts 20%–90% higher S in the melt during decompression than the MI trend (R2020). SolEx, with smaller relative differences from Sulfur_X at pressure between 400 and 250 MPa, cannot be applied to pressures higher than 400 MPa. And the relative difference between SolEx and R2020 increases rapidly to 80% from 250 to 100 MPa. Though the mismatch between Sulfur_X's prediction and R2020 is still non-negligible (we discussed the potential causes of this mismatch above), Sulfur_X fits the rehomogenized MI data better than other available models. Figure 8b compares the computed $kdS^{v/m}_{\text{combined}}$ from Sulfur_X to that from three other degassing models: SolEx, D-Compress, and R2020. Since these models employ different approaches to model sulfur degassing, we recalculated the apparent $kdS^{v/m}$ ($=w_S^{\text{vapor}}/w_S^{\text{melt}}$) using the modeled sulfur concentration in the vapor and in the melt to compare different models. The predicted $kdS^{v/m}_{\text{combined}}$ from Sulfur_X varies from 20 to 80 at pressures higher than 100 MPa, agreeing well with the partition coefficients (40–60) derived from R2020. It is not surprising that Sulfur_X results diverge from R2020 significantly in both Figures 8a and 8b during the last stage of degassing because most MIs were entrapped at pressures higher than 100 MPa, and so have no information to guide the empirical fit between 0 and 100 MPa. Although both the apparent $kdS^{v/m}$ from SolEx and D-Compress increase significantly at low pressure like Sulfur_X, they differ strongly at high pressure. For SolEx, the maximum initial pressure that can be used is 400 MPa, the highest experimental pressure of the calibration experiments (Lesne et al., 2011). Therefore, the model run starts with ~1,100 ppm CO_2 and 2,100 ppm S at 400 MPa. Between pressure of 400 and 100 MPa, the apparent $kdS^{v/m}$ from SolEx starts around 20 but rapidly decreases to <10, significantly lower than the $kdS^{v/m}_{\text{combined}}$ at the same pressure. This low apparent $kdS^{v/m}$ explains why SolEx predicts almost no sulfur degassing for Fuego until pressures lower than 100 MPa, when CO_2 is almost entirely degassed (Figure 1).

In contrast, the apparent $kdS^{v/m}$ from D-Compress appears to be at least one order of magnitude higher than the $kdS^{v/m}_{\text{combined}}$ from Sulfur_X and is responsible for the concave-up shape of the model CO_2-S degassing trend in Figure 1. It is important to note that D-Compress does not allow initial S concentration as a model input. Instead, fSO_2, fH_2S and fS_2 are calculated in the vapor based on the input of initial fO_2 (input as FMQ + 1.2), initial H_2O (4.5 wt.%), and initial CO_2 (3,300 ppm) in the system by gas equilibrium, fugacity laws and mass balance. The initial pressure was 650 MPa, and the temperature was 1,030°C. Such starting inputs give S concentration of

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~2,700 ppm in the melt, very close to the initial conditions run by Sulfur_X. However, D-Compress calculates 46 mol% S_2 together with 18 mol% SO_2 and 11 mol% H_2S in the initial vapor. Figure 8e shows that for most of the degassing path ($P > 50$ MPa), fSO_2 and fH_2S predicted by D-Compress are almost an order of magnitude higher than those predicted by Sulfur_X. The predicted high fSO_2, fH_2S, and fS_2 make S the most abundant volatile species in the vapor until the pressure drops to <100 MPa. Since the mass of S is conserved in the system, high sulfur contents in the vapor cause the rapid decrease of S in the melt and, thus, the high apparent $kdS^{v/m}$ shown in Figure 8b. The high total S in the vapor also results in low fH_2O (Figure 8e) at high pressure due to dilution of H_2O in the vapor phase. D-Compress solves for fSO_2, fH_2S, fS_2 in the vapor together with the fugacities of the other six gas species at each pressure step (Burgisser et al., 2015), and it is unclear what factor(s) caused the unreasonably high fSO_2, fH_2S, and fS_2.

We can also compare the fO_2 and S speciation evolution between Sulfur_X and D-Compress. The predicted SO_2/S_T from D-Compress increases dramatically as pressure decreases (Figure 8c), primarily due to the fast decrease of fS_2 in the vapor (Figure 8e). The predicted $S^{6+}/\Sigma S$ in the melt from D-Compress follows the changing $SO_2/(SO_2 + H_2S)$ in the vapor until 100 MPa. This is not surprising because D-Compress does not directly calculate the S speciation in the melt. Instead, it computes the concentrations of S^{2-} and S^{6+} in the melt separately using fH_2S and fSO_2 and their solubility laws (Burgisser et al., 2015). Therefore, D-Compress only considers fO_2 change due to reactions between different volatile species in the vapor and reaction between fO_2, Fe^{3+}, and Fe^{2+} in the melt. It ignores the S redox change when S degasses from melt to the vapor and its potential effects on $Fe^{3+}/\Sigma Fe$. However, our model shows that fO_2 in the system might be dominated by different processes at different degassing stages. The gas equilibrium for S species in the vapor (H_2S converting to SO_2 for Fuego case) plays an important role at low pressure, where most S has already been degassed. RxnII, S^{6+} degassing to SO_2, on the other hand, significantly affects $Fe^{3+}/\Sigma Fe$ in the melt, thus the fO_2, during high pressure degassing.

4.2. Magma Degassing in Mauna Kea Volcano and Comparison to Arc Magma Degassing

We applied Sulfur_X to the degassing of H_2O-poor and more reduced magmas from Hawaii with lower initial ΔFMQ and lower initial $S^{6+}/\Sigma S$ ratio (~0.1). This example provides a contrast to our earlier example of an H_2O-rich, high initial $S^{6+}/\Sigma S$ ratio (~0.7) arc magma from Fuego. For the Hawaiian degassing model, the initial H_2O and S concentrations are taken from the highest concentrations measured in Mauna Kea glasses (Brounce et al., 2017): 0.6 wt.% H_2O and 1,500 ppm S. The initial CO_2 of the undegassed magma is assumed to be 4,000 ppm, among the highest CO_2 concentrations measured in Kilauea Iki MIs after restoration using Raman spectroscopy and 1 atm reheating experiments (L. R. Moore et al., 2015; Tuohy et al., 2016). The initial fO_2 is FMQ + 0.5 to match the $Fe^{3+}/\Sigma Fe$ ratios in the least degassed Mauna Kea submarine glass (0.18), first measured by Brounce et al. (2017) and updated by Lerner (2020) using pre-edge centroid energy and an updated Mossbauer calibration (Zhang et al., 2018). The model temperature is fixed at 1,150-C. The measured $S^{6+}/\Sigma S$ associated with the least degassed sample from Mauna Kea (Brounce et al., 2017) is assumed for the initial $S^{6+}/\Sigma S$ in the melt (~0.1). Since none of the existing models can reproduce the glass speciation data at the given temperature, we used the MW20 model and modified the last constant to match the observed initial $Fe^{3+}/\Sigma Fe$ (0.18) and initial $S^{6+}/\Sigma S$ (0.1) in the pillow rim glasses from Mauna Kea. Crystallization is disabled in this run as the lower H_2O of Hawaiian magma results in no decompression-driven crystallization until the very end of the degassing path. We employed the VolatileCalc degassing model (Newman & Lowenstern, 2002) for these low-H_2O magmas. To stabilize the model run in the low pressure range, after 5 MPa, instead of using the computed $kdS^{v/m}_{\text{combined}}$, we assumed that $kdS^{v/m}_{\text{combined}}$ increases by 50 at each degassing step until 1 atm.

The model results are plotted in Figures 9 and 10. One of the critical observations from the model results for Hawaiian magma is the S evolution in the melt as a function of pressure (Figure 9a). In contrast to the prediction that Fuego magma degasses S from 500 to 100 MPa (dashed curves in Figure 9a), Sulfur_X predicts that S in Mauna Kea melt stays almost constant until the pressure drops to below 100 MPa (with an $S^{6+}/\Sigma S$ in the melt of 0.1). 90% of the total S degassing occurs in the last 45 MPa during magma ascent. This predicted shallow degassing of S is consistent with the semi-empirical model results from Gerlach (1986) and Lerner et al. (2021) that both S and H_2O degassing do not begin until the melts reach 100–200 m (2–5 MPa) beneath the surface. However, Sulfur_X predicts that S degasses before H_2O, which matches the S-H_2O trends defined by Mauna Kea glasses (Figure 10a; Brounce et al., 2017). The low-pressure (<40 MPa) of sulfur degassing is a direct result of the $kdS^{v/m}_{\text{combined}}$, a weighted average of the $kdS^{v/m}_{\text{RxnI}}$ and $kdS^{v/m}_{\text{RxnII}}$. Figure 9b shows that the calculated $kdS^{v/m}_{\text{RxnI}}$ is

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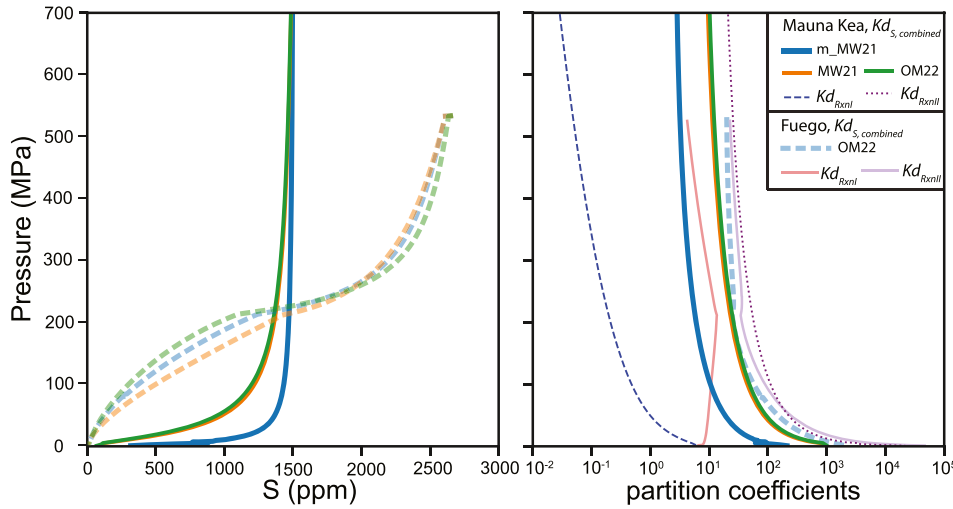

Figure 9. Comparisons between results of Fuego and Mauna Kea starting compositions. (a) Modeled S concentration in the melt and (b) $kdS^{v/m}_{\mathrm{RxnI}}$, $kdS^{v/m}_{\mathrm{RxnII}}$, and $kdS^{v/m}_{\mathrm{combined}}$ change as pressure decreases. In panel (a), thick solid curves represent results from Mauna Kea and thick-dashed curves are results from Fuego. Different colors represent different S-Fe speciation models used. Blue: modified S-Fe model for Mauna Kea, O'Neill and Mavrognese (2022) for Fuego; orange: Muth and Wallace (2021); green: Nash et al. (2019). In panel (b), blue and purple thin dashed curves are calculated $kdS^{v/m}_{\mathrm{RxnI}}$ and $kdS^{v/m}_{\mathrm{RxnII}}$ for Mauna Kea while thick solid curves are $kdS^{v/m}_{\mathrm{combined}}$ for Mauna Kea for different S-Fe models. Pink and light purple thin solid curves are calculated $kdS^{v/m}_{\mathrm{RxnI}}$, $kdS^{v/m}_{\mathrm{RxnII}}$ for Fuego while thick dashed curve is $kdS^{v/m}_{\mathrm{combined}}$ for Fuego by O'Neill and Mavrogenes (2022).

between 0.01 and 1 until the pressure drops to below 50 MPa, two orders of magnitude lower than the $kdS^{v/m}_{\mathrm{RxnI}}$ for Fuego magma. Since both H_2O mole fraction in the melt and H_2O fugacity can enhance $kdS^{v/m}_{\mathrm{RxnI}}$, the ~10 times lower H_2O concentration and five times lower H_2O fugacity of Mauna Kea magma likely causes the much lower predicted $kdS^{v/m}_{\mathrm{RxnI}}$ compared to Fuego magma at a given pressure. In contrast, $kdS^{v/m}_{\mathrm{RxnII}}$, which is not directly affected by H_2O in the system, is very similar for Mauna Kea and Fuego magmas (Figure 9b). However, the Mauna Kea magma has a low initial $S^{6+}/\Sigma S$ of around 0.1 (Figure 10c, Brounce et al., 2017). As a result, $kdS^{v/m}_{\mathrm{RxnI}}$ dominates the $kdS^{v/m}_{\mathrm{combined}}$ (Figure 9b). Since $kdS^{v/m}_{\mathrm{RxnI}}$ maintains low values (between 0.1 and 1) until the last 50 MPa, $kdS^{v/m}_{\mathrm{combined}}$ and S degassing are low for Hawaiian magmas until shallow pressure. In the meantime, the co-existing vapor has >0.9 SO_2/S_T due to the high absolute fO_2, temperature, and low fH_2O (Figure 10b). As a result, Fe^{3+} in the melt receives electrons from S^{2-} when H_2S^{2-} converts to $S^{4+}O_2$ and is reduced as degassing

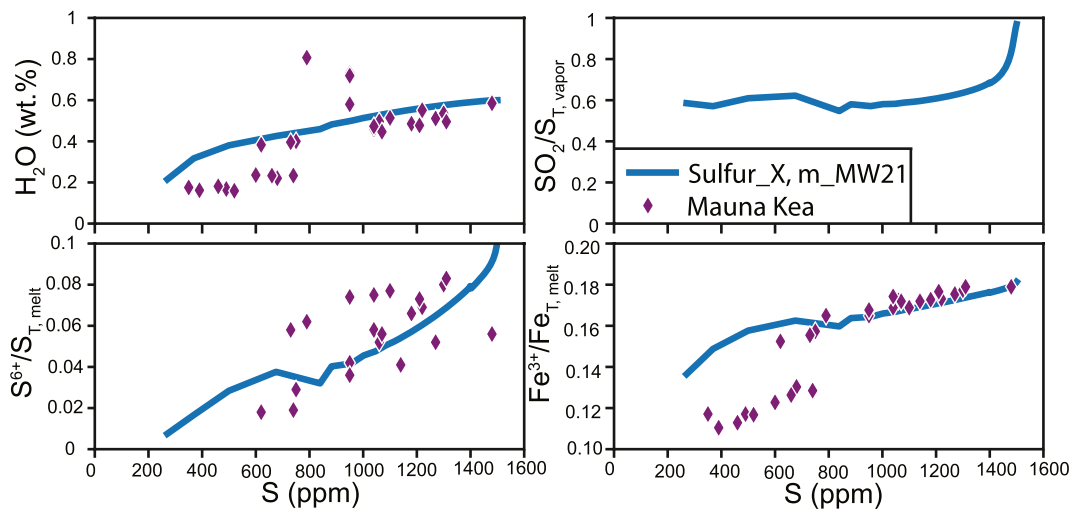

Figure 10. Comparisons of the measured (symbols) and modeled (dashed lines) S and Fe speciation changes during degassing for Mauna Kea (pillow rim glasses data from Brounce et al., 2017). Modeling details are described in the main text.

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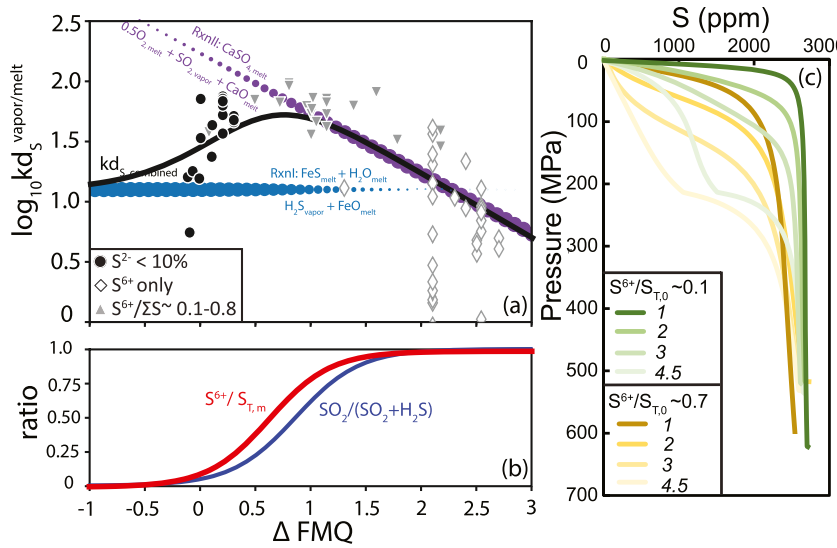

Figure 11. (a) Sulfur partition coefficients for RxnI (blue circles), RxnII (purple circles), and combined sulfur partition coefficient (black curve) and (b) S speciation in the melt (red curve) and vapor (blue curve) as a function of oxygen fugacity (relative to FMQ buffer). In panel (a), the experimental partition coefficients with different $S^{6+}/\Sigma S$ in the melt (as in Figure 2b) are plotted for comparison. (c) Modeled sulfur concentration for Fuego composition at 1,030°C as a function of pressure with low (0.1, green curves) and high (0.7, yellow curves) initial S^{6+}/S_T in the melt. Under each condition, S degassing was modeled with different initial water contents (4.5, 3, 2, 1 wt.% H_2O). Other initial conditions are the same. In panel (a), the $kd_{S,\text{combined}}^{\text{vapor/melt}}$ is calculated using Equation 11 in the text. The relative contribution of each reaction to $kd_{S,\text{combined}}^{\text{vapor/melt}}$ is indicated by the sizes of the symbols (as in Equation 11). The *kdS* were calculated at a pressure of 200 MPa, temperature of 1,030°C, and melt composition of mafic Fuego melt inclusions with 3.85 wt.% H_2O and ~58 mol% of H_2O in the vapor.

proceeds (Figure 10d). The reduction of Fe drives a decrease in fO_2, which increases the H_2S fraction in the vapor (Figure 10b) and requires more of RxnI's contribution. The effect from RxnII appears to be very small given the small amount of S^{6+} in the melt and its low efficiency in transferring electrons compared to the equilibrium between H_2S and SO_2 in the vapor. Therefore, predicted $S^{6+}/\Sigma S$ and $Fe^{3+}/\Sigma Fe$ in Mauna Kea magma continuously decreased, in contrast to the more complex behavior displayed by Fuego magma. As a result of decreasing fO_2, SO_2/S_T in the vapor decreases from >0.95 to ~0.6 (Figure 10b) and maintains around 0.6 as water fugacity continues to fall at low pressure. The predicted SO_2/S_T in the vapor at 1 atm (~0.6), however, is lower than that from the gas measurements at the surface (Figure 10b), which is between 0.9 and 1 (Gerlach, 1986). Figure 10 also shows the comparison between the results of Sulfur_X and the H_2O, S, $S^{6+}/\Sigma S$, and $Fe^{3+}/\Sigma Fe$ measured in submarine glasses from Mauna Kea (Brounce et al., 2017). Figure 10a shows that the modeled melt H_2O-S evolution for Mauna Kea passes through the submarine glass data, showing significant S degassing (>50%) when little H_2O degasses (<0.1 wt.% H_2O). This is in contrast to the predictions from Gerlach (1986) and Lerner et al. (2021) that S only begins to degas when H_2O degasses significantly. Figures 10c and 10d show that the modeled $S^{6+}/\Sigma S$ and $Fe^{3+}/\Sigma Fe$ decrease as S degasses, similar to that measured in the Mauna Kea glasses.

It is important to note that although the initial fO_2 relative to the FMQ buffer of Hawaiian magma is lower than that of Fuego, the equivalent absolute log fO_2 value calculated for the initial melt composition, pressure, and temperature, is around −7.7 (Kress & Carmichael, 1991), higher than that calculated for Fuego magma (~−8.7). Since $kdS_{\text{RxnII}}^{v/m}$ depend on absolute fO_2 as an input, its values are not appreciably lower as might be expected for more reduced magma than Fuego. Temperature might not be the main factor either. Figures S4 and S5 in Supporting Information S1 show that an increase of temperature from 1,050°C to 1,150°C causes relatively small change in $kdS_{\text{RxnI}}^{v/m}$ and $kdS_{\text{RxnII}}^{v/m}$. We emphasize here that the key impact of initial fO_2 on S degassing in Hawaiian magmas is through the low initial $S^{6+}/\Sigma S$ in the melt and the consequent reduction during degassing. Degassing paths calculated using N19 and MW21 with initial $S^{6+}/\Sigma S$ of 0.45 and 0.47 are plotted in Figure 9 for direct comparisons. Given the much higher initial $S^{6+}/\Sigma S$ in the melt predicted by these two models, $kdS_{\text{combined}}^{v/m}$ is much higher than that of the modified model (Figure 9b). As a result, S starts degassing at a higher pressure, ~200 MPa (Figure 9b). The predicted $S^{6+}/\Sigma S$ and $Fe^{3+}/\Sigma Fe$ also diverge from the measured glass data when starting with higher $S^{6+}/\Sigma S$ in the melt (Figure S7 in Supporting Information S1).

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5. Implications

5.1. General Application to Volcanic Systems

Our model results show that in Fuego magma, a typical H_2O-rich (4.5 wt.%) arc magma, with high initial $S^{6+}/\Sigma S$ (~0.7), ~70% of the sulfur in the melt is lost by degassing at pressures above 200 MPa (dashed curves in Figure 9a). In contrast, in Mauna Kea magma, an H_2O-poor (<1 wt.%) oceanic island basalt with low initial $S^{6+}/\Sigma S$ (~0.1), almost no sulfur degasses until shallow pressures (<45 MPa, dashed curve in Figure 9a). The contrasting modeled sulfur degassing paths challenge current degassing models (e.g., SolEx, MP04) that predict sulfur loss only at pressures lower than 150 MPa (e.g., J. G. Moore & Fabbi, 1971; J. G. Moore & Schilling, 1973; Webster et al., 2011). Despite all of the experiments used for calibrating Sulfur_X that demonstrates the preference of sulfur for the vapor phase to varying extents from 70 to 500 MPa, the SolEx and MP04 models are still widely applied to interpret MI data (e.g., Ferguson et al., 2016; Robidoux et al., 2018) and high-T volcanic gas data (e.g., Aiuppa et al., 2017; de Moor et al., 2016; Robidoux et al., 2017). Forward modeling by Sulfur_X shows that S degassing varies depending on the evolution of $kdS^{v/m}_{\mathrm{RxnI}}$, and $kdS^{v/m}_{\mathrm{RxnII}}$ as functions of temperature, pressure, fO_2, H_2O content, and melt composition, which affect each partition coefficient differently. For example, with other conditions the same, increasing fO_2 decreases $kdS^{v/m}_{\mathrm{RxnII}}$ but does not affect $kdS^{fl/m}_{\mathrm{RxnI}}$. Low H_2O in the system significantly suppresses RxnI, but it does not affect RxnII. Moreover, the combined partition coefficient also depends on S speciation in the melt (Equation 11). S degassing from reduced magmas with mostly S^{2-} in the melt would be dominated by $kdS^{v/m}_{\mathrm{RxnI}}$, whereas S degassing from oxidized magmas with mostly S^{6+} would be dictated by $kdS^{v/m}_{\mathrm{RxnII}}$. With decreasing fO_2, $kdS^{v/m}_{\mathrm{RxnII}}$ increases but S^{6+} decreases, thereby decreasing the contribution of $kdS^{v/m}_{\mathrm{RxnII}}$ to the $kdS^{v/m}_{\mathrm{combined}}$ (illustrated with the decrease in size of the purple dots in Figure 11a). $kdS^{v/m}_{\mathrm{RxnI}}$ shows no variation with fO_2 because O_2 is not involved in this reaction. And yet, as fO_2 increases, the amount of S^{2-} in the melt decreases, thereby decreasing the contribution of $kdS^{v/m}_{\mathrm{RxnI}}$ to the $kdS^{v/m}_{\mathrm{combined}}$. At intermediate fO_2, the combined effect of $kdS^{fl/m}_{\mathrm{RxnI}}$ and $kdS^{fl/m}_{\mathrm{RxnII}}$ as a function of fO_2 leads to a peak in $kdS^{v/m}_{\mathrm{combined}}$, capturing the maximum $kdS^{v/m}$ displayed by experimental data (Figure 11a). This maximum $kdS^{v/m}_{\mathrm{combined}}$ leads to enhanced partitioning of S into the vapor, and thus more efficient degassing of S as observed for Fuego. This maximum in S partitioning into the vapor corresponds to the "minimum" in sulfur solubility in the melt that has been commonly observed in previous experiments (e.g., Carroll & Rutherford, 1985; Fincham et al., 1954) and recently modeled by Hughes et al. (2022) in the simple S-O-silicate system (see also discussion of sulfur solubility "minimum" in O'Neill, 2021).

It is also important to consider the effects of H_2O fugacity. At high H_2O fugacity, $kdS^{v/m}_{\mathrm{RxnI}}$ is increased. Therefore, arc magmas like Fuego with intermediate fO_2 and high H_2O in the system can receive contributions from both reactions to $kdS^{v/m}_{\mathrm{combined}}$, and thus generally degas sulfur efficiently even at high pressures. In contrast, for relatively reduced and H_2O-poor OIBs, the lack of S^{6+} minimizes the contribution from $kdS^{v/m}_{\mathrm{RxnII}}$. At the same time, low fH_2O decreases $kdS^{v/m}_{\mathrm{RxnI}}$. Thus, partitioning of S to vapor is minimal, and only occurs significantly as P drops below 50 MPa, due to the $-\ln P$ effect on $kdS^{v/m}_{\mathrm{RxnI}}$ and $kdS^{v/m}_{\mathrm{RxnII}}$.

Sulfur degassing can be even more complex than described above for Fuego and Hawaii. For example, reduced magma does not necessarily have low $kdS^{v/m}_{\mathrm{RxnII}}$ if the calculated absolute fO_2 is not low, as shown in Mauna Kea submarine glasses. The absolute fO_2 is not only related to $Fe^{3+}/\Sigma Fe$ in the melt but also pressure, temperature, and melt composition (Kress & Carmichael, 1991). Dry magma may have low $kdS^{v/m}_{\mathrm{RxnI}}$ due to low water in the system, but $kdS^{v/m}_{\mathrm{RxnII}}$ is not directly affected by H_2O and can still be significant, as shown in the case of Mauna Kea (Figure 10b). Therefore, if a dry system had more S^{6+} in the melt than magmas from Hawaii, a significant amount of S could still be degassed earlier than the Hawaiian cases (e.g., Figure 9a when the N19 or MW21 models are applied). The general effects of initial H_2O and $S^{6+}/\Sigma S$ in the melt on sulfur degassing as a function of pressure are illustrated in Figure 11c. With high initial $S^{6+}/\Sigma S$ (~0.7) in the melt, significant sulfur degassing happens at pressures higher than 200 MPa, regardless of the H_2O contents, although higher H_2O contents result in much earlier degassing. On the contrary, with low initial $S^{6+}/\Sigma S$ (~0.1), S begins to degas deep only in wet magma (4.5–3 wt.% H_2O), similar to that in the magmas with high initial $S^{6+}/\Sigma S$. In this case, a decrease in H_2O concentration dramatically decreases S degassing depth. In addition, Sulfur_X shows that Fe and S redox can change in complex ways during decompression depending on the initial $S^{6+}/\Sigma S$ in the melt and other parameters. Such feedback between S-Fe speciation in the melt, S speciation in the vapor and S degassing is captured for the first time by Sulfur_X.

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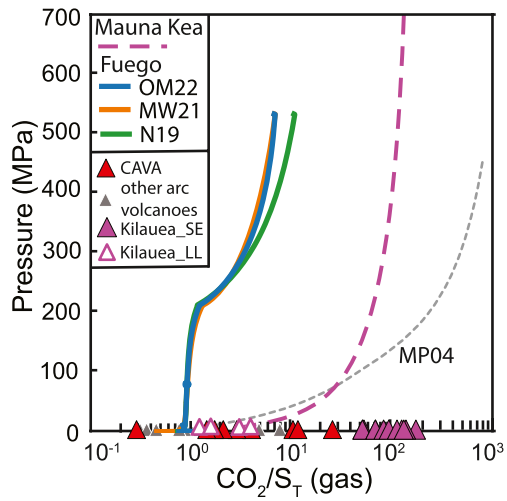

Figure 12. CO_2/S_T ratios in vapor co-existing with degassing melt as a function of pressure for Fuego and Kilauea from Sulfur_X (thick lines). Plotted for comparison are modeled CO_2/S_T ratios in Fuego vapor by Moretti and Papale (2004) model (MP04); measured CO_2/S_T from some central American volcanoes (Aiuppa et al., 2014, 2018; Moussallam et al., 2017), other arc volcanoes worldwide (Fischer, 2008), around Halemaumau lava lake in 1918–1919 (Kilauea_LL, Gerlach, 1980, 1986) and from Kilauea summit emissions from 1995 to 1999 (Kilauea_SE, Gerlach et al., 2002).

5.2. Application to Volcanic Gas Compositions

Sulfur_X also predicts the compositional evolution of co-existing vapor during magma decompression, which can be particularly useful in interpreting the CO_2/S_T molar ratio measured in high-*T* volcanic gases. First of all, CO_2/S_T ratios measured in high-*T* volcanic gases are often used to infer degassing depths in arc volcanic systems (Aiuppa et al., 2016, 2017; Burton et al., 2007; Maarten de Moor et al., 2016). For example, a significant increase from 0.69 to 2.2 to a peak of 9 in CO_2/S_T ratio in volcanic gases from Villarrica and Turrialba volcanoes in the weeks before eruptions were interpreted as a surge of CO_2-enriched gas coming from ~30 MPa to 200–250 MPa for Villarrica and Turrialba, respectively (Aiuppa et al., 2017; de Moor et al., 2016) using the MP04 model (Moretti & Papale, 2004). In Figure 12, the predicted CO_2/S_T ratios in the co-existing vapor for Fuego are plotted against pressure. The predicted CO_2/S_T ratios between 0.5 and 10 agree well with the gas compositions collected at the surface from several Central American volcanoes (CAVA), and other arc volcanoes worldwide. For comparison, the CO_2/S_T ratios in the co-existing vapor for Fuego modeled by MP04 (Moretti & Papale, 2004), the most commonly used degassing model by the volcanic gas community, are also plotted as a function of pressure in Figure 12. In contrast to the orders of magnitude increase in the predicted CO_2/S_T ratios from within pressure between 1 atm and 200 MPa by MP04, Sulfur_X predicts that CO_2/S_T ratios for Fuego gases are <5 at pressures lower than 300 MPa and only increase up to 10 at pressures above 500 MPa. Therefore, if the peak CO_2/S_T (~10) observed in high-*T* volcanic gases from arc volcanoes before eruptions truly represents a surge of deep magma and gas, results from Sulfur_X for Fuego suggest that magma injection could happen much deeper than previously inferred (>500 vs. ~10 MPa by MP04).

In addition to being a depth signature, Aiuppa et al. (2014, 2017) proposed that the time-averaged CO_2/S_T ratio in volcanic gas can be a source feature, with some volcanoes having intrinsically higher ratios than others. Indeed, if the magma is completely degassed at 1 atm, Sulfur_X predicts that the CO_2/S_T molar ratio in the gas in equilibrium with the magma at 1 atm will be the same as the initial CO_2/S_T in the parental magma by closed-system degassing. For example, Figure 12 shows that the modeled CO_2/S_T ratio of gas emitted at 1 atm for Fuego is around 0.89 starting with 3,300 ppm CO_2, and 1.62 starting with 6,000 ppm CO_2. For Kilauea, in comparison, the value is 1.8. These numbers are very close to the initial molar ratio of the CO_2/S_T in the melt. However, Figure 12 also shows that CO_2/S_T ratios from passive degassing of most arc volcanoes are between 1 and 5, higher than the modeled CO_2/S_T ratio of gas emitted at 1 atm for Fuego. One possibility is that most arc volcanoes have higher CO_2/S_T than Fuego. The other possibility is that although collected at the surface, instead of equilibrating at 1 atm, the measured gas represents mixtures of gas derived from different depths, accounting for the variations of the measured CO_2/S_T in the volcanic gases. Lastly, the predicted CO_2/S_T molar ratios for Mauna Kea volcano by Sulfur_X are about one order of magnitude higher than that for Fuego magma and remain greater than 50 along most of the decompression path, which results in little sulfur degassing at pressure >50 MPa (see Figure 9a). As pressure drops below 50 MPa, the predicted CO_2/S_T ratios for Mauna Kea drop rapidly from 50 to ~1.8 at the surface. Gas compositions collected around Halemaumau lava lake of Kilauea volcano in 1918–1919 (Gerlach, 1980, 1986) with CO_2/S_T molar ratios of 1.3–4.1 and directly measured from Kilauea summit emissions by 3 sets of summit correlation spectrometer (COSPEC) traverse from 1995 to 1999 (Gerlach et al., 2002) with CO_2/S_T molar ratios between 50 and 200 are both plotted in Figure 12. Comparing the gas compositions to the modeled results suggests that gases collected around the Halemaumau lava lake released from pressures below 10 MPa from a parental magma enriched in CO_2 (type I magma in Gerlach, 1986), while the very high CO_2/S_T ratios from the 1995–1999 summit emissions suggest that the gases could have separated from the magma at pressures higher than 100 MPa.

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6. Conclusions

1. Our analysis of published sulfur partitioning experiments for basalt and basaltic andesite melts shows that sulfur degassing can be ascribed to two different reactions: S^{2-} in the melt degassing to H_2S in the vapor (RxnI), and S^{6+} in the melt to SO_2 in the vapor (RxnII). Guided by a broad set of experiments, Sulfur_X models S degassing from ascending magmas by calculating separate partition coefficients for each sulfur degassing reaction. This approach also allows Sulfur_X to account for pressure and temperature effects using thermodynamic formulations instead of empirically extrapolating the results of low pressure experiments to high pressure (>200 MPa) where sulfur partitioning experiments are scarce. Sulfur_X then empirically combines the two partition coefficients based on the sulfur species in the melt to formulate a bulk sulfur partition coefficient, which becomes the key parameter in Sulfur_X for modeling sulfur degassing. Sulfur_X also predicts the redox evolution of sulfur and iron in the system using redox budget conservation.
2. Applying Sulfur_X to Fuego and Mauna Kea shows that a typical wet (4.5 wt.% H_2O), oxidized arc magma with high initial $Fe^{3+}/\Sigma Fe$ and $S^{6+}/\Sigma S$ ratios can lose 2/3 of its sulfur by degassing at pressures above 200 MPa (e.g., Fuego) whereas dry (<1 wt.%) and more reduced oceanic island basalts with low initial $S^{6+}/\Sigma S$ ratios degas little sulfur until shallow pressures (<45 MPa, e.g., Hawaii). Such pressure-sulfur degassing relationships calculated using Sulfur_X bear directly on the interpretation of CO_2/S gas precursors to volcanic eruptions, of obvious relevancy to developing eruption forecasts.
3. Sulfur _X explains the complex effects of fO_2 on sulfur degassing that are observed in experimental data. Sulfur_X predicts that the effect of both *kdS* and $S^{6+}/\Sigma S$ as a function of fO_2 leads to a peak in the $kdS^{v/m}_{\text{combined}}$ at intermediate fO_2, and so a maximum in sulfur degassing in arc magmas with high initial $S^{6+}/\Sigma S$ (>0.5) in the melt. Furthermore, comparison between the model results of the two case studies shows the importance of both H_2O concentration in the melt and H_2O fugacity in increasing $kdS^{fl/m}_{\text{RxnI}}$, which is important in sulfur degassing in magmas dominated by S^{2-} in the melt.
4. Sulfur_X shows that the two sulfur degassing reactions (RxnI, and RxnII), and thus the process of sulfur degassing, are related to pressure, temperature, melt composition, H_2O fugacity, and oxygen fugacity of the system. Therefore, it shows the importance of complete MI data sets that have measured H_2O, S, reconstructed CO_2 concentration, and $Fe^{3+}/\Sigma Fe$ and $S^{6+}/\Sigma S$ ratios. It also highlights the need for new experiments at pressures higher than 300 MPa with controlled *P-T-*fO_2 conditions.

Data Availability Statement

Python code of Sulfur_X is published in Zenodo (DOI: https://doi.org/10.5281/zenodo.7662826) and can be accessed at https://github.com/sdecho/Sulfur_X.git.

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